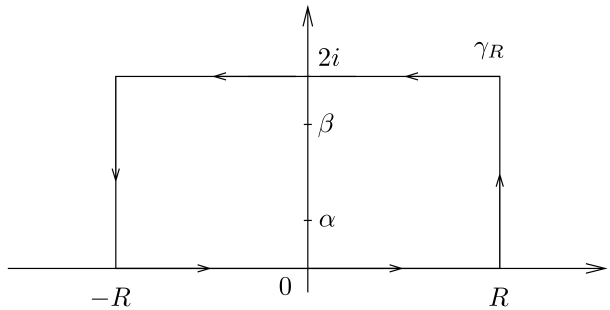

Figure 3. The contour γ_R in Example 3

and note that the denominator of f vanishes precisely when $e^{\pi z} = -e^{-\pi z}$, that is, when $e^{2\pi z} = -1$. In other words, the only poles of f inside the rectangle are at the points $\alpha = i/2$ and $\beta = 3i/2$. To find the residue of f at α, we note that

$$\begin{aligned}(z-\alpha)f(z) &= e^{-2\pi iz\xi}\frac{2(z-\alpha)}{e^{\pi z}+e^{-\pi z}}\\ &= 2e^{-2\pi iz\xi}e^{\pi z}\frac{(z-\alpha)}{e^{2\pi z}-e^{2\pi\alpha}}.\end{aligned}$$

We recognize on the right the reciprocal of the difference quotient for the function $e^{2\pi z}$ at $z = \alpha$. Therefore

$$\lim_{z\to\alpha}(z-\alpha)f(z) = 2e^{-2\pi i\alpha\xi}e^{\pi\alpha}\frac{1}{2\pi e^{2\pi\alpha}} = \frac{e^{\pi\xi}}{\pi i},$$

which shows that f has a simple pole at α with residue $e^{\pi\xi}/(\pi i)$. Similarly, we find that f has a simple pole at β with residue $-e^{3\pi\xi}/(\pi i)$.

We dispense with the integrals of f on the vertical sides by showing that they go to zero as R tends to infinity. Indeed, if $z = R + iy$ with $0 \leq y \leq 2$, then

$$|e^{-2\pi iz\xi}| \leq e^{4\pi|\xi|},$$

and

$$\begin{aligned} |\cosh \pi z| &= \left| \frac{e^{\pi z} + e^{-\pi z}}{2} \right| \\ &\geq \frac{1}{2} \left| |e^{\pi z}| - |e^{-\pi z}| \right| \\ &\geq \frac{1}{2}(e^{\pi R} - e^{-\pi R}) \\ &\to \infty \qquad \text{as } R \to \infty, \end{aligned}$$

which shows that the integral over the vertical segment on the right goes to 0 as $R \to \infty$. A similar argument shows that the integral of f over the vertical segment on the left also goes to 0 as $R \to \infty$. Finally, we see that if I denotes the integral we wish to calculate, then the integral of f over the top side of the rectangle (with the orientation from right to left) is simply $-e^{4\pi\xi}I$ where we have used the fact that $\cosh \pi\zeta$ is periodic with period $2i$. In the limit as R tends to infinity, the residue formula gives

$$\begin{aligned} I - e^{4\pi\xi} I &= 2\pi i \left(\frac{e^{\pi\xi}}{\pi i} - \frac{e^{3\pi\xi}}{\pi i} \right) \\ &= -2e^{2\pi\xi}(e^{\pi\xi} - e^{-\pi\xi}), \end{aligned}$$

and since $1 - e^{4\pi\xi} = -e^{2\pi\xi}(e^{2\pi\xi} - e^{-2\pi\xi})$, we find that

$$I = 2\frac{e^{\pi\xi} - e^{-\pi\xi}}{e^{2\pi\xi} - e^{-2\pi\xi}} = 2\frac{e^{\pi\xi} - e^{-\pi\xi}}{(e^{\pi\xi} - e^{-\pi\xi})(e^{\pi\xi} + e^{-\pi\xi})} = \frac{2}{e^{\pi\xi} + e^{-\pi\xi}} = \frac{1}{\cosh \pi\xi}$$

as claimed.

A similar argument actually establishes the following formula:

$$\int_{-\infty}^{\infty} e^{-2\pi i x\xi} \frac{\sin \pi a}{\cosh \pi x + \cos \pi a}\, dx = \frac{2 \sinh 2\pi a\xi}{\sinh 2\pi\xi}$$

whenever $0 < a < 1$, and where $\sinh z = (e^z - e^{-z})/2$. We have proved above the particular case $a = 1/2$. This identity can be used to determine an explicit formula for the Poisson kernel for the strip (see Problem 3 in Chapter 5 of Book I), or to prove the sum of two squares theorem, as we shall see in Chapter 10.

3 Singularities and meromorphic functions

Returning to Section 1, we see that we have described the analytical character of a function near a pole. We now turn our attention to the other types of isolated singularities.

Let f be a function holomorphic in an open set Ω except possibly at one point z_0 in Ω. If we can define f at z_0 in such a way that f becomes holomorphic in all of Ω, we say that z_0 is a **removable** singularity for f.

Theorem 3.1 (Riemann's theorem on removable singularities) *Suppose that f is holomorphic in an open set Ω except possibly at a point z_0 in Ω. If f is bounded on $\Omega - \{z_0\}$, then z_0 is a removable singularity.*

Proof. Since the problem is local we may consider a small disc D centered at z_0 and whose closure is contained in Ω. Let C denote the boundary circle of that disc with the usual positive orientation. We shall prove that if $z \in D$ and $z \neq z_0$, then under the assumptions of the theorem we have

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} \, d\zeta. \tag{4}$$

Since an application of Theorem 5.4 in the previous chapter proves that the right-hand side of equation (4) defines a holomorphic function on *all* of D that agrees with $f(z)$ when $z \neq z_0$, this give us the desired extension.

To prove formula (4) we fix $z \in D$ with $z \neq z_0$ and use the familiar toy contour illustrated in Figure 4.

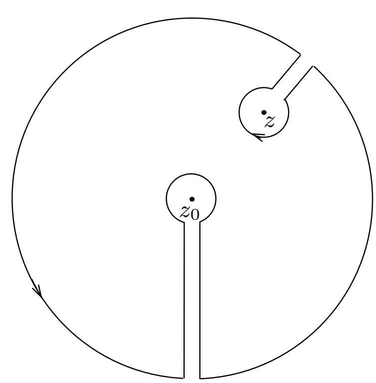

Figure 4. The multiple keyhole contour in the proof of Riemann's theorem

The multiple keyhole avoids the two points z and z_0. Letting the sides of the corridors get closer to each other, and finally overlap, in the limit we get a cancellation:

$$\int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = 0\,,$$

where γ_ϵ and γ'_ϵ are small circles of radius ϵ with negative orientation and centered at z and z_0 respectively. Copying the argument used in the proof of the Cauchy integral formula in Section 4 of Chapter 2, we find that

$$\int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = -2\pi i f(z).$$

For the second integral, we use the assumption that f is bounded and that since ϵ is small, ζ stays away from z, and therefore

$$\left| \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta \right| \leq C\epsilon.$$

Letting ϵ tend to 0 proves our contention and concludes the proof of the extension formula (4).

Surprisingly, we may deduce from Riemann's theorem a characterization of poles in terms of the behavior of the function in a neighborhood of a singularity.

Corollary 3.2 *Suppose that f has an isolated singularity at the point z_0. Then z_0 is a pole of f if and only if $|f(z)| \to \infty$ as $z \to z_0$.*

Proof. If z_0 is a pole, then we know that $1/f$ has a zero at z_0, and therefore $|f(z)| \to \infty$ as $z \to z_0$. Conversely, suppose that this condition holds. Then $1/f$ is bounded near z_0, and in fact $1/|f(z)| \to 0$ as $z \to z_0$. Therefore, $1/f$ has a removable singularity at z_0 and must vanish there. This proves the converse, namely that z_0 is a pole.

Isolated singularities belong to one of three categories:

- Removable singularities (f bounded near z_0)
- Pole singularities ($|f(z)| \to \infty$ as $z \to z_0$)
- Essential singularities.

By default, any singularity that is not removable or a pole is defined to be an **essential singularity**. For example, the function $e^{1/z}$ discussed at the very beginning of Section 1 has an essential singularity at

0. We already observed the wild behavior of this function near the origin. Contrary to the controlled behavior of a holomorphic function near a removable singularity or a pole, it is typical for a holomorphic function to behave erratically near an essential singularity. The next theorem clarifies this.

Theorem 3.3 (Casorati-Weierstrass) *Suppose f is holomorphic in the punctured disc $D_r(z_0) - \{z_0\}$ and has an essential singularity at z_0. Then, the image of $D_r(z_0) - \{z_0\}$ under f is dense in the complex plane.*

Proof. We argue by contradiction. Assume that the range of f is not dense, so that there exists $w \in \mathbb{C}$ and $\delta > 0$ such that

$$|f(z) - w| > \delta \qquad \text{for all } z \in D_r(z_0) - \{z_0\}.$$

We may therefore define a new function on $D_r(z_0) - \{z_0\}$ by

$$g(z) = \frac{1}{f(z) - w},$$

which is holomorphic on the punctured disc and bounded by $1/\delta$. Hence g has a removable singularity at z_0 by Theorem 3.1. If $g(z_0) \neq 0$, then $f(z) - w$ is holomorphic at z_0, which contradicts the assumption that z_0 is an essential singularity. In the case that $g(z_0) = 0$, then $f(z) - w$ has a pole at z_0 also contradicting the nature of the singularity at z_0. The proof is complete.

In fact, Picard proved a much stronger result. He showed that under the hypothesis of the above theorem, the function f takes on every complex value infinitely many times with at most one exception. Although we shall not give a proof of this remarkable result, a simpler version of it will follow from our study of entire functions in a later chapter. See Exercise 11 in Chapter 5.

We now turn to functions with only isolated singularities that are poles. A function f on an open set Ω is **meromorphic** if there exists a sequence of points $\{z_0, z_1, z_2, \ldots\}$ that has no limit points in Ω, and such that

(i) the function f is holomorphic in $\Omega - \{z_0, z_1, z_2, \ldots\}$, and

(ii) f has poles at the points $\{z_0, z_1, z_2, \ldots\}$.

It is also useful to discuss functions that are meromorphic in the extended complex plane. If a function is holomorphic for all large values of

z, we can describe its behavior at infinity using the tripartite distinction we have used to classify singularities at finite values of z. Thus, if f is holomorphic for all large values of z, we consider $F(z) = f(1/z)$, which is now holomorphic in a deleted neighborhood of the origin. We say that f has a **pole at infinity** if F has a pole at the origin. Similarly, we can speak of f having an **essential singularity at infinity**, or a **removable singularity** (hence holomorphic) **at infinity** in terms of the corresponding behavior of F at 0. A meromorphic function in the complex plane that is either holomorphic at infinity or has a pole at infinity is said to be **meromorphic in the extended complex plane**.

At this stage we return to the principle mentioned at the beginning of the chapter. Here we can see it in its simplest form.

Theorem 3.4 *The meromorphic functions in the extended complex plane are the rational functions.*

Proof. Suppose that f is meromorphic in the extended plane. Then $f(1/z)$ has either a pole or a removable singularity at 0, and in either case it must be holomorphic in a deleted neighborhood of the origin, so that the function f can have only finitely many poles in the plane, say at $z_1, \ldots, z_n$. The idea is to subtract from f its principal parts at all its poles including the one at infinity. Near each pole $z_k \in \mathbb{C}$ we can write

$$f(z) = f_k(z) + g_k(z),$$

where $f_k(z)$ is the principal part of f at z_k and g_k is holomorphic in a (full) neighborhood of z_k. In particular, f_k is a polynomial in $1/(z - z_k)$. Similarly, we can write

$$f(1/z) = \tilde{f}_\infty(z) + \tilde{g}_\infty(z),$$

where $\tilde{g}_\infty$ is holomorphic in a neighborhood of the origin and $\tilde{f}_\infty$ is the principal part of $f(1/z)$ at 0, that is, a polynomial in $1/z$. Finally, let $f_\infty(z) = \tilde{f}_\infty(1/z)$.

We contend that the function $H = f - f_\infty - \sum_{k=1}^n f_k$ is entire and bounded. Indeed, near the pole z_k we subtracted the principal part of f so that the function H has a removable singularity there. Also, $H(1/z)$ is bounded for z near 0 since we subtracted the principal part of the pole at ∞. This proves our contention, and by Liouville's theorem we conclude that H is constant. From the definition of H, we find that f is a rational function, as was to be shown.

Note that as a consequence, a rational function is determined up to a multiplicative constant by prescribing the locations and multiplicities of its zeros and poles.

The Riemann sphere

The extended complex plane, which consists of $\mathbb{C}$ and the point at infinity, has a convenient geometric interpretation, which we briefly discuss here.

Consider the Euclidean space $\mathbb{R}^3$ with coordinates (X, Y, Z) where the XY-plane is identified with $\mathbb{C}$. We denote by $\mathbb{S}$ the sphere centered at $(0, 0, 1/2)$ and of radius $1/2$; this sphere is of unit diameter and lies on top of the origin of the complex plane as pictured in Figure 5. Also, we let $\mathcal{N} = (0, 0, 1)$ denote the north pole of the sphere.

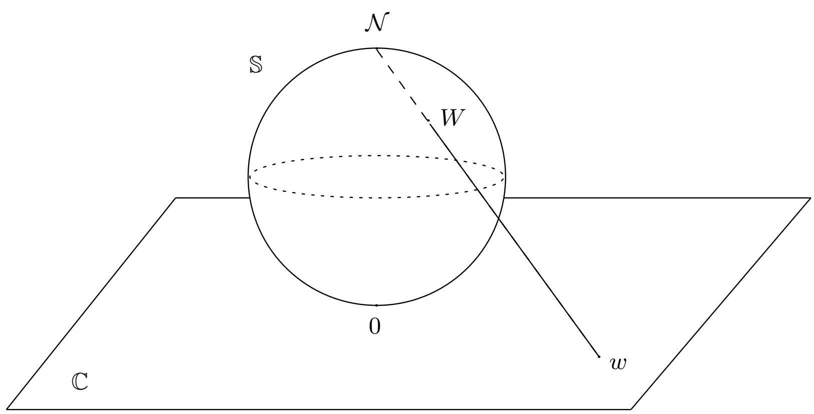

Figure 5. The Riemann sphere $\mathbb{S}$ and stereographic projection

Given any point $W = (X, Y, Z)$ on $\mathbb{S}$ different from the north pole, the line joining $\mathcal{N}$ and W intersects the XY-plane in a single point which we denote by $w = x + iy$; w is called the **stereographic projection** of W (see Figure 5). Conversely, given any point w in $\mathbb{C}$, the line joining $\mathcal{N}$ and $w = (x, y, 0)$ intersects the sphere at $\mathcal{N}$ and another point, which we call W. This geometric construction gives a bijective correspondence between points on the punctured sphere $\mathbb{S} - \{\mathcal{N}\}$ and the complex plane; it is described analytically by the formulas

$$x = \frac{X}{1 - Z} \quad \text{and} \quad y = \frac{Y}{1 - Z},$$

giving w in terms of W, and

$$X = \frac{x}{x^2 + y^2 + 1}, \quad Y = \frac{y}{x^2 + y^2 + 1}, \quad \text{and} \quad Z = \frac{x^2 + y^2}{x^2 + y^2 + 1}$$

giving W in terms of w. Intuitively, we have wrapped the complex plane onto the punctured sphere $\mathbb{S} - \{\mathcal{N}\}$.

As the point w goes to infinity in $\mathbb{C}$ (in the sense that $|w| \to \infty$) the corresponding point W on $\mathbb{S}$ comes arbitrarily close to $\mathcal{N}$. This simple observation makes $\mathcal{N}$ a natural candidate for the so-called "point at infinity." Identifying infinity with the point $\mathcal{N}$ on $\mathbb{S}$, we see that the extended complex plane can be visualized as the full two-dimensional sphere $\mathbb{S}$; this is the **Riemann sphere**. Since this construction takes the unbounded set $\mathbb{C}$ into the compact set $\mathbb{S}$ by adding one point, the Riemann sphere is sometimes called the **one-point compactification** of $\mathbb{C}$.

An important consequence of this interpretation is the following: although the point at infinity required special attention when considered separately from $\mathbb{C}$, it now finds itself on equal footing with all other points on $\mathbb{S}$. In particular, a meromorphic function on the extended complex plane can be thought of as a map from $\mathbb{S}$ to itself, where the image of a pole is now a tractable point on $\mathbb{S}$, namely $\mathcal{N}$. For these reasons (and others) the Riemann sphere provides good geometrical insight into the structure of $\mathbb{C}$ as well as the theory of meromorphic functions.

4 The argument principle and applications

We anticipate our discussion of the logarithm (in Section 6) with a few comments. In general, the function $\log f(z)$ is "multiple-valued" because it cannot be defined unambiguously on the set where $f(z) \neq 0$. However it is to be defined, it must equal $\log|f(z)| + i \arg f(z)$, where $\log|f(z)|$ is the usual real-variable logarithm of the positive quantity $|f(z)|$ (and hence is defined unambiguously), while $\arg f(z)$ is some determination of the argument (up to an additive integral multiple of 2π). Note that in any case, the derivative of $\log f(z)$ is $f'(z)/f(z)$ which is single-valued, and the integral

$$\int_\gamma \frac{f'(z)}{f(z)}\, dz$$

can be interpreted as the change in the argument of f as z traverses the curve γ. Moreover, assuming the curve is closed, this change of argument is determined entirely by the zeros and poles of f inside γ. We now formulate this fact as a precise theorem.

We begin with the observation that while the additivity formula

$$\log(f_1 f_2) = \log f_1 + \log f_2$$

fails in general (as we shall see below), the additivity can be restored to the corresponding derivatives. This is confirmed by the following observation:

$$\frac{(f_1 f_2)'}{f_1 f_2} = \frac{f_1' f_2 + f_1 f_2'}{f_1 f_2} = \frac{f_1'}{f_1} + \frac{f_2'}{f_2},$$

which generalizes to

$$\frac{\left(\prod_{k=1}^N f_k\right)'}{\prod_{k=1}^N f_k} = \sum_{k=1}^N \frac{f_k'}{f_k}.$$

We apply this formula as follows. If f is holomorphic and has a zero of order n at z_0, we can write

$$f(z) = (z - z_0)^n g(z)\,,$$

where g is holomorphic and nowhere vanishing in a neighborhood of z_0, and therefore

$$\frac{f'(z)}{f(z)} = \frac{n}{z - z_0} + G(z)$$

where $G(z) = g'(z)/g(z)$. The conclusion is that if f has a zero of order n at z_0, then f'/f has a simple pole with residue n at z_0. Observe that a similar fact also holds if f has a pole of order n at z_0, that is, if $f(z) = (z - z_0)^{-n} h(z)$. Then

$$\frac{f'(z)}{f(z)} = \frac{-n}{z - z_0} + H(z).$$

Therefore, if f is meromorphic, the function f'/f will have simple poles at the zeros and poles of f, and the residue is simply the order of the zero of f or the negative of the order of the pole of f. As a result, an application of the residue formula gives the following theorem.

Theorem 4.1 (Argument principle) *Suppose f is meromorphic in an open set containing a circle C and its interior. If f has no poles and never vanishes on C, then*

$$\frac{1}{2\pi i}\int_C \frac{f'(z)}{f(z)}\,dz = \textit{(number of zeros of } f \textit{ inside } C\textit{) minus}$$
$$\textit{(number of poles of } f \textit{ inside } C\textit{),}$$

where the zeros and poles are counted with their multiplicities.

Corollary 4.2 *The above theorem holds for toy contours.*

As an application of the argument principle, we shall prove three theorems of interest in the general theory. The first, Rouché's theorem, is in some sense a continuity statement. It says that a holomorphic function can be perturbed slightly without changing the number of its zeros. Then, we prove the open mapping theorem, which states that holomorphic functions map open sets to open sets, an important property that again shows the special nature of holomorphic functions. Finally, the maximum modulus principle is reminiscent of (and in fact implies) the same property for harmonic functions: a non-constant holomorphic function on an open set Ω cannot attain its maximum in the interior of Ω.

Theorem 4.3 (Rouché's theorem) *Suppose that f and g are holomorphic in an open set containing a circle C and its interior. If*

$$|f(z)| > |g(z)| \quad \text{for all } z \in C,$$

then f and $f+g$ have the same number of zeros inside the circle C.

Proof. For $t \in [0,1]$ define

$$f_t(z) = f(z) + tg(z)$$

so that $f_0 = f$ and $f_1 = f + g$. Let n_t denote the number of zeros of f_t inside the circle counted with multiplicities, so that in particular, n_t is an integer. The condition $|f(z)| > |g(z)|$ for $z \in C$ clearly implies that f_t has no zeros on the circle, and the argument principle implies

$$n_t = \frac{1}{2\pi i}\int_C \frac{f_t'(z)}{f_t(z)}\, dz.$$

To prove that n_t is constant, it suffices to show that it is a continuous function of t. Then we could argue that if n_t were not constant, the intermediate value theorem would guarantee the existence of some $t_0 \in [0,1]$ with n_{t_0} not integral, contradicting the fact that $n_t \in \mathbb{Z}$ for all t.

To prove the continuity of n_t, we observe that $f_t'(z)/f_t(z)$ is jointly continuous for $t \in [0,1]$ and $z \in C$. This joint continuity follows from the fact that it holds for both the numerator and denominator, and our assumptions guarantee that $f_t(z)$ does not vanish on C. Hence n_t is integer-valued and continuous, and it must be constant. We conclude that $n_0 = n_1$, which is Rouché's theorem.

We now come to an important geometric property of holomorphic functions that arises when we consider them as mappings (that is, mapping regions in the complex plane to the complex plane).

A mapping is said to be **open** if it maps open sets to open sets.

Theorem 4.4 (Open mapping theorem) *If f is holomorphic and non-constant in a region Ω, then f is open.*

Proof. Let w_0 belong to the image of f, say $w_0 = f(z_0)$. We must prove that all points w near w_0 also belong to the image of f.

Define $g(z) = f(z) - w$ and write

$$\begin{aligned} g(z) &= (f(z) - w_0) + (w_0 - w) \\ &= F(z) + G(z). \end{aligned}$$

Now choose $\delta > 0$ such that the disc $|z - z_0| \leq \delta$ is contained in Ω and $f(z) \neq w_0$ on the circle $|z - z_0| = \delta$. We then select $\epsilon > 0$ so that we have $|f(z) - w_0| \geq \epsilon$ on the circle $|z - z_0| = \delta$. Now if $|w - w_0| < \epsilon$ we have $|F(z)| > |G(z)|$ on the circle $|z - z_0| = \delta$, and by Rouché's theorem we conclude that $g = F + G$ has a zero inside the circle since F has one.

The next result pertains to the size of a holomorphic function. We shall refer to the **maximum** of a holomorphic function f in an open set Ω as the maximum of its absolute value $|f|$ in Ω.

Theorem 4.5 (Maximum modulus principle) *If f is a non-constant holomorphic function in a region Ω, then f cannot attain a maximum in Ω.*

Proof. Suppose that f did attain a maximum at z_0. Since f is holomorphic it is an open mapping, and therefore, if $D \subset \Omega$ is a small disc centered at z_0, its image $f(D)$ is open and contains $f(z_0)$. This proves that there are points in $z \in D$ such that $|f(z)| > |f(z_0)|$, a contradiction.

Corollary 4.6 *Suppose that Ω is a region with compact closure $\overline{\Omega}$. If f is holomorphic on Ω and continuous on $\overline{\Omega}$ then*

$$\sup_{z \in \Omega} |f(z)| \leq \sup_{z \in \overline{\Omega} - \Omega} |f(z)|.$$

In fact, since $f(z)$ is continuous on the compact set $\overline{\Omega}$, then $|f(z)|$ attains its maximum in $\overline{\Omega}$; but this cannot be in Ω if f is non-constant. If f is constant, the conclusion is trivial.

Remark. The hypothesis that $\overline{\Omega}$ is compact (that is, bounded) is essential for the conclusion. We give an example related to considerations that we will take up in Chapter 4. Let Ω be the open first quadrant, bounded by the positive half-line $x \geq 0$ and the corresponding imaginary line $y \geq 0$. Consider $F(z) = e^{-iz^2}$. Then F is entire and clearly continuous on $\overline{\Omega}$. Moreover $|F(z)| = 1$ on the two boundary lines $z = x$ and $z = iy$. However, $F(z)$ is unbounded in Ω, since for example, we have $F(z) = e^{r^2}$ if $z = r\sqrt{i} = re^{i\pi/4}$.

5 Homotopies and simply connected domains

The key to the general form of Cauchy's theorem, as well as the analysis of multiple-valued functions, is to understand in what regions we can define the primitive of a given holomorphic function. Note the relevance to the study of the logarithm, which arises as a primitive of $1/z$. The question is not just a local one, but is also global in nature. Its elucidation requires the notion of homotopy, and the resulting idea of simple-connectivity.

Let γ_0 and γ_1 be two curves in an open set Ω with common end-points. So if $\gamma_0(t)$ and $\gamma_1(t)$ are two parametrizations defined on $[a, b]$, we have

$$\gamma_0(a) = \gamma_1(a) = \alpha \quad \text{and} \quad \gamma_0(b) = \gamma_1(b) = \beta.$$

These two curves are said to be **homotopic in** Ω if for each $0 \leq s \leq 1$ there exists a curve $\gamma_s \subset \Omega$, parametrized by $\gamma_s(t)$ defined on $[a, b]$, such that for every s

$$\gamma_s(a) = \alpha \quad \text{and} \quad \gamma_s(b) = \beta,$$

and for all $t \in [a, b]$

$$\gamma_s(t)|_{s=0} = \gamma_0(t) \quad \text{and} \quad \gamma_s(t)|_{s=1} = \gamma_1(t).$$

Moreover, $\gamma_s(t)$ should be jointly continuous in $s \in [0, 1]$ and $t \in [a, b]$.

Loosely speaking, two curves are homotopic if one curve can be deformed into the other by a continuous transformation without ever leaving Ω (Figure 6).

Theorem 5.1 *If f is holomorphic in Ω, then*

$$\int_{\gamma_0} f(z)\,dz = \int_{\gamma_1} f(z)\,dz$$

whenever the two curves γ_0 and γ_1 are homotopic in Ω.

Proof. The key to the proof lies in showing that if two curves are close to each other and have the same end-points, then the integrals over these curves are equal. Recall that by definition, the function $F(s, t) = \gamma_s(t)$ is continuous on $[0, 1] \times [a, b]$. In particular, since the image of F, which we

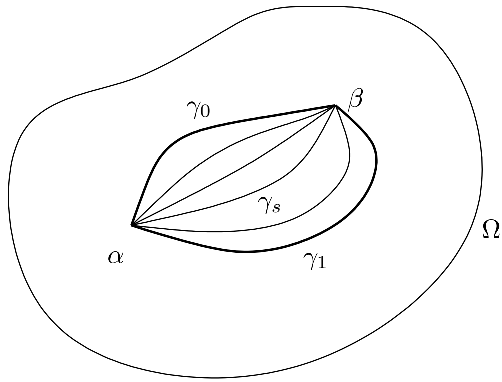

Figure 6. Homotopy of curves

denote by K, is compact, there exists $\epsilon > 0$ such that every disc of radius 3ϵ centered at a point in the image of F is completely contained in Ω. If not, for every $\ell \geq 0$, there exist points $z_\ell \in K$ and w_ℓ in the complement of Ω such that $|z_\ell - w_\ell| < 1/\ell$. By compactness of K, there exists a subsequence of $\{z_\ell\}$, say $\{z_{\ell_k}\}$, that converges to a point $z \in K \subset \Omega$. By construction, we must have $w_{\ell_k} \to z$ as well, and since $\{w_\ell\}$ lies in the complement of Ω which is closed, we must have $z \in \Omega^c$ as well. This is a contradiction.

Having found an ϵ with the desired property, we may, by the uniform continuity of F, select δ so that

$$\sup_{t \in [a,b]} |\gamma_{s_1}(t) - \gamma_{s_2}(t)| < \epsilon \qquad \text{whenever } |s_1 - s_2| < \delta.$$

Fix s_1 and s_2 with $|s_1 - s_2| < \delta$. We then choose discs $\{D_0, \ldots, D_n\}$ of radius 2ϵ, and consecutive points $\{z_0, \ldots, z_{n+1}\}$ on γ_{s_1} and $\{w_0, \ldots, w_{n+1}\}$ on γ_{s_2} such that the union of these discs covers both curves, and

$$z_i, z_{i+1}, w_i, w_{i+1} \in D_i.$$

The situation is illustrated in Figure 7.

Also, we choose $z_0 = w_0$ as the beginning end-point of the curves and $z_{n+1} = w_{n+1}$ as the common end-point. On each disc D_i, let F_i denote a primitive of f (Theorem 2.1, Chapter 2). On the intersection of D_i and D_{i+1}, F_i and F_{i+1} are two primitives of the same function, so they must differ by a constant, say c_i. Therefore

$$F_{i+1}(z_{i+1}) - F_i(z_{i+1}) = F_{i+1}(w_{i+1}) - F_i(w_{i+1}),$$

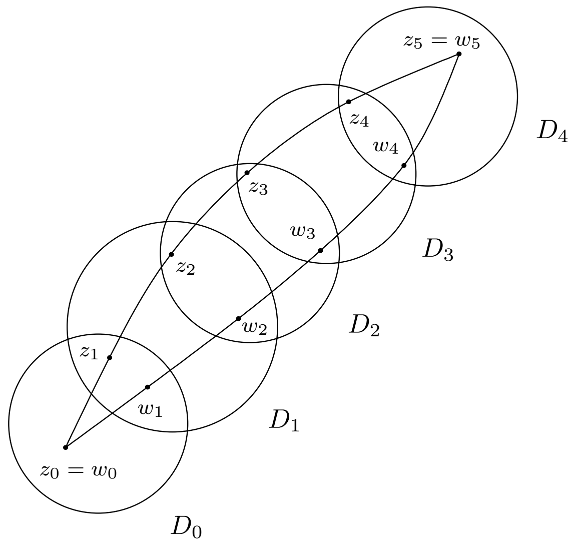

Figure 7. Covering two nearby curves with discs

hence

$$F_{i+1}(z_{i+1}) - F_{i+1}(w_{i+1}) = F_i(z_{i+1}) - F_i(w_{i+1}). \tag{5}$$

This implies

$$\begin{aligned}\int_{\gamma_{s_1}} f - \int_{\gamma_{s_2}} f &= \sum_{i=0}^{n}[F_i(z_{i+1}) - F_i(z_i)] - \sum_{i=0}^{n}[F_i(w_{i+1}) - F_i(w_i)] \\ &= \sum_{i=0}^{n} F_i(z_{i+1}) - F_i(w_{i+1}) - (F_i(z_i) - F_i(w_i)) \\ &= F_n(z_{n+1}) - F_n(w_{n+1}) - (F_0(z_0) - F_0(w_0)),\end{aligned}$$

because of the cancellations due to (5). Since γ_{s_1} and γ_{s_2} have the same beginning and end point, we have proved that

$$\int_{\gamma_{s_1}} f = \int_{\gamma_{s_2}} f.$$

By subdividing $[0,1]$ into subintervals $[s_i, s_{i+1}]$ of length less than δ, we may go from γ_0 to γ_1 by finitely many applications of the above argument, and the theorem is proved.

A region Ω in the complex plane is **simply connected** if any two pair of curves in Ω with the same end-points are homotopic.

EXAMPLE 1. A disc D is simply connected. In fact, if $\gamma_0(t)$ and $\gamma_1(t)$ are two curves lying in D, we can define $\gamma_s(t)$ by

$$\gamma_s(t) = (1-s)\gamma_0(t) + s\gamma_1(t).$$

Note that if $0 \leq s \leq 1$, then for each t, the point $\gamma_s(t)$ is on the segment joining $\gamma_0(t)$ and $\gamma_1(t)$, and so is in D. The same argument works if D is replaced by a rectangle, or more generally by any open convex set. (See Exercise 21.)

EXAMPLE 2. The slit plane $\Omega = \mathbb{C} - \{(-\infty, 0]\}$ is simply connected. For a pair of curves γ_0 and γ_1 in Ω, we write $\gamma_j(t) = r_j(t)e^{i\theta_j(t)}$ $(j = 0, 1)$ with $r_j(t)$ continuous and strictly positive, and $\theta_j(t)$ continuous with $|\theta_j(t)| < \pi$. Then, we can define $\gamma_s(t)$ as $r_s(t)e^{i\theta_s(t)}$ where

$$r_s(t) = (1-s)r_0(t) + sr_1(t) \quad \text{and} \quad \theta_s(t) = (1-s)\theta_0(t) + s\theta_1(t).$$

We then have $\gamma_s(t) \in \Omega$ whenever $0 \leq s \leq 1$.

EXAMPLE 3. With some effort one can show that the interior of a toy contour is simply connected. This requires that we divide the interior into several subregions. A general form of the argument is given in Exercise 4.

EXAMPLE 4. In contrast with the previous examples, the punctured plane $\mathbb{C} - \{0\}$ is not simply connected. Intuitively, consider two curves with the origin between them. It is impossible to continuously pass from one curve to the other without going over 0. A rigorous proof of this fact requires one further result, and will be given shortly.

Theorem 5.2 *Any holomorphic function in a simply connected domain has a primitive.*

Proof. Fix a point z_0 in Ω and define

$$F(z) = \int_\gamma f(w)\, dw$$

where the integral is taken over any curve in Ω joining z_0 to z. This definition is independent of the curve chosen, since Ω is simply connected, and if $\tilde{\gamma}$ is another curve in Ω joining z_0 and z, we would have

$$\int_\gamma f(w)\,dw = \int_{\tilde{\gamma}} f(w)\,dw$$

by Theorem 5.1. Now we can write

$$F(z+h) - F(z) = \int_\eta f(w)\,dw$$

where η is the line segment joining z and $z+h$. Arguing as in the proof of Theorem 2.1 in Chapter 2, we find that

$$\lim_{h\to 0} \frac{F(z+h)-F(z)}{h} = f(z).$$

As a result, we obtain the following version of Cauchy's theorem.

Corollary 5.3 *If f is holomorphic in the simply connected region Ω, then*

$$\int_\gamma f(z)\,dz = 0$$

for any closed curve γ in Ω.

This is immediate from the existence of a primitive.

The fact that the punctured plane is not simply connected now follows rigorously from the observation that the integral of $1/z$ over the unit circle is $2\pi i$, and not 0.

6 The complex logarithm

Suppose we wish to define the logarithm of a non-zero complex number. If $z = re^{i\theta}$, and we want the logarithm to be the inverse to the exponential, then it is natural to set

$$\log z = \log r + i\theta.$$

Here and below, we use the convention that $\log r$ denotes the *standard*[1] logarithm of the positive number r. The trouble with the above definition is that θ is unique only up to an integer multiple of 2π. However,

[1] By the standard logarithm, we mean the natural logarithm of positive numbers that appears in elementary calculus.

for given z we can fix a choice of θ, and if z varies only a little, this determines the corresponding choice of θ uniquely (assuming we require that θ varies continuously with z). Thus "locally" we can give an unambiguous definition of the logarithm, but this will not work "globally." For example, if z starts at 1, and then winds around the origin and returns to 1, the logarithm does not return to its original value, but rather differs by an integer multiple of $2\pi i$, and therefore is not "single-valued." To make sense of the logarithm as a single-valued function, we must restrict the set on which we define it. This is the so-called choice of a **branch** or **sheet** of the logarithm.

Our discussion of simply connected domains given above leads to a natural global definition of a branch of the logarithm function.

Theorem 6.1 *Suppose that Ω is simply connected with $1 \in \Omega$, and $0 \notin \Omega$. Then in Ω there is a branch of the logarithm $F(z) = \log_\Omega(z)$ so that*

(i) *F is holomorphic in Ω,*

(ii) *$e^{F(z)} = z$ for all $z \in \Omega$,*

(iii) *$F(r) = \log r$ whenever r is a real number and near 1.*

In other words, each branch $\log_\Omega(z)$ is an extension of the standard logarithm defined for positive numbers.

Proof. We shall construct F as a primitive of the function $1/z$. Since $0 \notin \Omega$, the function $f(z) = 1/z$ is holomorphic in Ω. We define

$$\log_\Omega(z) = F(z) = \int_\gamma f(w)\, dw\,,$$

where γ is any curve in Ω connecting 1 to z. Since Ω is simply connected, this definition does not depend on the path chosen. Arguing as in the proof of Theorem 5.2, we find that F is holomorphic and $F'(z) = 1/z$ for all $z \in \Omega$. This proves (i). To prove (ii), it suffices to show that $ze^{-F(z)} = 1$. For that, we differentiate the left-hand side, obtaining

$$\frac{d}{dz}\left(ze^{-F(z)}\right) = e^{-F(z)} - zF'(z)e^{-F(z)} = (1 - zF'(z))e^{-F(z)} = 0.$$

Since Ω is connected we conclude, by Corollary 3.4 in Chapter 1, that $ze^{-F(z)}$ is constant. Evaluating this expression at $z = 1$, and noting that $F(1) = 0$, we find that this constant must be 1.

Finally, if r is real and close to 1 we can choose as a path from 1 to r a line segment on the real axis so that

$$F(r) = \int_1^r \frac{dx}{x} = \log r,$$

by the usual formula for the standard logarithm. This completes the proof of the theorem.

For example, in the slit plane $\Omega = \mathbb{C} - \{(-\infty, 0]\}$ we have the **principal branch** of the logarithm

$$\log z = \log r + i\theta$$

where $z = re^{i\theta}$ with $|\theta| < \pi$. (Here we drop the subscript Ω, and write simply $\log z$.) To prove this, we use the path of integration γ shown in Figure 8.

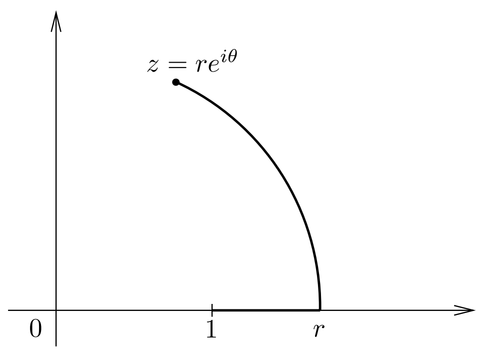

Figure 8. Path of integration for the principal branch of the logarithm

If $z = re^{i\theta}$ with $|\theta| < \pi$, the path consists of the line segment from 1 to r and the arc η from r to z. Then

$$\begin{aligned}\log z &= \int_1^r \frac{dx}{x} + \int_\eta \frac{dw}{w} \\ &= \log r + \int_0^\theta \frac{ire^{it}}{re^{it}}\, dt \\ &= \log r + i\theta.\end{aligned}$$

An important observation is that in general

$$\log(z_1 z_2) \neq \log z_1 + \log z_2.$$

For example, if $z_1 = e^{2\pi i/3} = z_2$, then for the principal branch of the logarithm, we have

$$\log z_1 = \log z_2 = \frac{2\pi i}{3},$$

and since $z_1 z_2 = e^{-2\pi i/3}$ we have

$$-\frac{2\pi i}{3} = \log(z_1 z_2) \neq \log z_1 + \log z_2.$$

Finally, for the principal branch of the logarithm the following Taylor expansion holds:

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots = -\sum_{n=1}^{\infty}(-1)^n \frac{z^n}{n} \quad \text{for } |z| < 1. \tag{6}$$

Indeed, the derivative of both sides equals $1/(1+z)$, so that they differ by a constant. Since both sides are equal to 0 at $z = 0$ this constant must be 0, and we have proved the desired Taylor expansion.

Having defined a logarithm on a simply connected domain, we can now define the powers z^α for any $\alpha \in \mathbb{C}$. If Ω is simply connected with $1 \in \Omega$ and $0 \notin \Omega$, we choose the branch of the logarithm with $\log 1 = 0$ as above, and define

$$z^\alpha = e^{\alpha \log z}.$$

Note that $1^\alpha = 1$, and that if $\alpha = 1/n$, then

$$(z^{1/n})^n = \prod_{k=1}^{n} e^{\frac{1}{n}\log z} = e^{\sum_{k=1}^{n}\frac{1}{n}\log z} = e^{\frac{n}{n}\log z} = e^{\log z} = z.$$

We know that every non-zero complex number w can be written as $w = e^z$. A generalization of this fact is given in the next theorem, which discusses the existence of $\log f(z)$ whenever f does not vanish.

Theorem 6.2 *If f is a nowhere vanishing holomorphic function in a simply connected region Ω, then there exists a holomorphic function g on Ω such that*

$$f(z) = e^{g(z)}.$$

The function $g(z)$ in the theorem can be denoted by $\log f(z)$, and determines a "branch" of that logarithm.

Proof. Fix a point z_0 in Ω, and define a function

$$g(z) = \int_\gamma \frac{f'(w)}{f(w)}\, dw + c_0,$$

where γ is any path in Ω connecting z_0 to z, and c_0 is a complex number so that $e^{c_0} = f(z_0)$. This definition is independent of the path γ since Ω is simply connected. Arguing as in the proof of Theorem 2.1, Chapter 2, we find that g is holomorphic with

$$g'(z) = \frac{f'(z)}{f(z)},$$

and a simple calculation gives

$$\frac{d}{dz}\left(f(z)e^{-g(z)}\right) = 0\,,$$

so that $f(z)e^{-g(z)}$ is constant. Evaluating this expression at z_0 we find $f(z_0)e^{-c_0} = 1$, so that $f(z) = e^{g(z)}$ for all $z \in \Omega$, and the proof is complete.

7 Fourier series and harmonic functions

In Chapter 4 we shall describe some interesting connections between complex function theory and Fourier analysis on the real line. The motivation for this study comes in part from the simple and direct relation between Fourier series on the circle and power series expansions of holomorphic functions in the disc, which we now investigate.

Suppose that f is holomorphic in a disc $D_R(z_0)$, so that f has a power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

that converges in that disc.

Theorem 7.1 *The coefficients of the power series expansion of f are given by*

$$a_n = \frac{1}{2\pi r^n}\int_0^{2\pi} f(z_0 + re^{i\theta})e^{-in\theta}\,d\theta$$

for all $n \geq 0$ and $0 < r < R$. Moreover,

$$0 = \frac{1}{2\pi r^n}\int_0^{2\pi} f(z_0 + re^{i\theta})e^{-in\theta}\,d\theta$$

whenever $n < 0$.

Proof. Since $f^{(n)}(z_0) = a_n n!$, the Cauchy integral formula gives

$$a_n = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \, d\zeta \,,$$

where γ is a circle of radius $0 < r < R$ centered at z_0 and with the positive orientation. Choosing $\zeta = z_0 + re^{i\theta}$ for the parametrization of this circle, we find that for $n \geq 0$

$$\begin{aligned} a_n &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{(z_0 + re^{i\theta} - z_0)^{n+1}} rie^{i\theta} \, d\theta \\ &= \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-i(n+1)\theta} e^{i\theta} \, d\theta \\ &= \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta} \, d\theta. \end{aligned}$$

Finally, even when $n < 0$, our calculation shows that we still have the identity

$$\frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta} \, d\theta = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \, d\zeta.$$

Since $-n > 0$, the function $f(\zeta)(\zeta - z_0)^{-n-1}$ is holomorphic in the disc, and by Cauchy's theorem the last integral vanishes.

The interpretation of this theorem is as follows. Consider $f(z_0 + re^{i\theta})$ as the restriction to the circle of a holomorphic function on the closure of a disc centered at z_0 with radius r. Then its Fourier coefficients vanish if $n < 0$, while those for $n \geq 0$ are equal (up to a factor of r^n) to coefficients of the power series of the holomorphic function f. The property of the vanishing of the Fourier coefficients for $n < 0$ reveals another special characteristic of holomorphic functions (and in particular their restrictions to any circle).

Next, since $a_0 = f(z_0)$, we obtain the following corollary.

Corollary 7.2 (Mean-value property) *If f is holomorphic in a disc $D_R(z_0)$, then*

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) \, d\theta, \qquad \textit{for any } 0 < r < R.$$

Taking the real parts of both sides, we obtain the following consequence.

Corollary 7.3 *If f is holomorphic in a disc $D_R(z_0)$, and $u = \mathrm{Re}(f)$, then*

$$u(z_0) = \frac{1}{2\pi}\int_0^{2\pi} u(z_0 + re^{i\theta})\, d\theta, \qquad \text{for any } 0 < r < R.$$

Recall that u is harmonic whenever f is holomorphic, and in fact, the above corollary is a property enjoyed by every harmonic function in the disc $D_R(z_0)$. This follows from Exercise 12 in Chapter 2, which shows that every harmonic function in a disc is the real part of a holomorphic function in that disc.

8 Exercises

1. Using Euler's formula

$$\sin \pi z = \frac{e^{i\pi z} - e^{-i\pi z}}{2i},$$

show that the complex zeros of $\sin \pi z$ are exactly at the integers, and that they are each of order 1.

Calculate the residue of $1/\sin \pi z$ at $z = n \in \mathbb{Z}$.

2. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{dx}{1 + x^4}.$$

Where are the poles of $1/(1 + z^4)$?

3. Show that

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + a^2}\, dx = \pi \frac{e^{-a}}{a}, \qquad \text{for } a > 0.$$

4. Show that

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + a^2}\, dx = \pi e^{-a}, \qquad \text{for all } a > 0.$$

5. Use contour integration to show that

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x\xi}}{(1 + x^2)^2}\, dx = \frac{\pi}{2}(1 + 2\pi|\xi|)e^{-2\pi|\xi|}$$

for all ξ real.

6. Show that

$$\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^{n+1}} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \pi.$$

7. Prove that

$$\int_0^{2\pi} \frac{d\theta}{(a+\cos\theta)^2} = \frac{2\pi a}{(a^2-1)^{3/2}}, \qquad \text{whenever } a > 1.$$

8. Prove that

$$\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{\sqrt{a^2-b^2}}$$

if $a > |b|$ and $a, b \in \mathbb{R}$.

9. Show that

$$\int_0^1 \log(\sin \pi x)\, dx = -\log 2.$$

[Hint: Use the contour shown in Figure 9.]

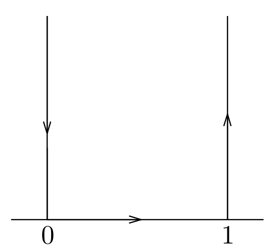

Figure 9. Contour in Exercise 9

10. Show that if $a > 0$, then

$$\int_0^{\infty} \frac{\log x}{x^2+a^2}\, dx = \frac{\pi}{2a} \log a.$$

[Hint: Use the contour in Figure 10.]

11. Show that if $|a| < 1$, then

$$\int_0^{2\pi} \log|1 - ae^{i\theta}|\, d\theta = 0.$$

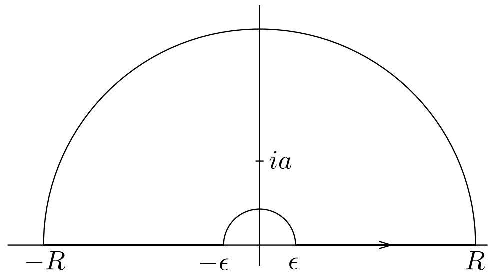

Figure 10. Contour in Exercise 10

Then, prove that the above result remains true if we assume only that $|a| \leq 1$.

12. Suppose u is not an integer. Prove that

$$\sum_{n=-\infty}^{\infty} \frac{1}{(u+n)^2} = \frac{\pi^2}{(\sin \pi u)^2}$$

by integrating

$$f(z) = \frac{\pi \cot \pi z}{(u+z)^2}$$

over the circle $|z| = R_N = N + 1/2$ (N integral, $N \geq |u|$), adding the residues of f inside the circle, and letting N tend to infinity.
Note. Two other derivations of this identity, using Fourier series, were given in Book I.

13. Suppose $f(z)$ is holomorphic in a punctured disc $D_r(z_0) - \{z_0\}$. Suppose also that

$$|f(z)| \leq A|z - z_0|^{-1+\epsilon}$$

for some $\epsilon > 0$, and all z near z_0. Show that the singularity of f at z_0 is removable.

14. Prove that all entire functions that are also injective take the form $f(z) = az + b$ with $a, b \in \mathbb{C}$, and $a \neq 0$.
[Hint: Apply the Casorati-Weierstrass theorem to $f(1/z)$.]

15. Use the Cauchy inequalities or the maximum modulus principle to solve the following problems:

(a) Prove that if f is an entire function that satisfies

$$\sup_{|z|=R} |f(z)| \leq AR^k + B$$

for all $R > 0$, and for some integer $k \geq 0$ and some constants $A, B > 0$, then f is a polynomial of degree $\leq k$.

(b) Show that if f is holomorphic in the unit disc, is bounded, and converges uniformly to zero in the sector $\theta < \arg z < \varphi$ as $|z| \to 1$, then $f = 0$.

(c) Let $w_1, \ldots, w_n$ be points on the unit circle in the complex plane. Prove that there exists a point z on the unit circle such that the product of the distances from z to the points w_j, $1 \leq j \leq n$, is at least 1. Conclude that there exists a point w on the unit circle such that the product of the distances from w to the points w_j, $1 \leq j \leq n$, is exactly equal to 1.

(d) Show that if the real part of an entire function f is bounded, then f is constant.

16. Suppose f and g are holomorphic in a region containing the disc $|z| \leq 1$. Suppose that f has a simple zero at $z = 0$ and vanishes nowhere else in $|z| \leq 1$. Let

$$f_\epsilon(z) = f(z) + \epsilon g(z).$$

Show that if ϵ is sufficiently small, then

(a) $f_\epsilon(z)$ has a unique zero in $|z| \leq 1$, and

(b) if z_ϵ is this zero, the mapping $\epsilon \mapsto z_\epsilon$ is continuous.

17. Let f be non-constant and holomorphic in an open set containing the closed unit disc.

(a) Show that if $|f(z)| = 1$ whenever $|z| = 1$, then the image of f contains the unit disc. [Hint: One must show that $f(z) = w_0$ has a root for every $w_0 \in \mathbb{D}$. To do this, it suffices to show that $f(z) = 0$ has a root (why?). Use the maximum modulus principle to conclude.]

(b) If $|f(z)| \geq 1$ whenever $|z| = 1$ and there exists a point $z_0 \in \mathbb{D}$ such that $|f(z_0)| < 1$, then the image of f contains the unit disc.

18. Give another proof of the Cauchy integral formula

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta$$

using homotopy of curves.

[Hint: Deform the circle C to a small circle centered at z, and note that the quotient $(f(\zeta) - f(z))/(\zeta - z)$ is bounded.]

19. Prove the maximum principle for harmonic functions, that is:

(a) If u is a non-constant real-valued harmonic function in a region Ω, then u cannot attain a maximum (or a minimum) in Ω.

(b) Suppose that Ω is a region with compact closure $\overline{\Omega}$. If u is harmonic in Ω and continuous in $\overline{\Omega}$, then

$$\sup_{z\in\Omega}|u(z)| \leq \sup_{z\in\overline{\Omega}-\Omega}|u(z)|.$$

[Hint: To prove the first part, assume that u attains a local maximum at z_0. Let f be holomorphic near z_0 with $u = \mathrm{Re}(f)$, and show that f is not open. The second part follows directly from the first.]

20. This exercise shows how the mean square convergence dominates the uniform convergence of analytic functions. If U is an open subset of $\mathbb{C}$ we use the notation

$$\|f\|_{L^2(U)} = \left(\int_U |f(z)|^2\, dxdy\right)^{1/2}$$

for the mean square norm, and

$$\|f\|_{L^\infty(U)} = \sup_{z\in U}|f(z)|$$

for the sup norm.

(a) If f is holomorphic in a neighborhood of the disc $D_r(z_0)$, show that for any $0 < s < r$ there exists a constant $C > 0$ (which depends on s and r) such that

$$\|f\|_{L^\infty(D_s(z_0))} \leq C\|f\|_{L^2(D_r(z_0))}\,.$$

(b) Prove that if $\{f_n\}$ is a Cauchy sequence of holomorphic functions in the mean square norm $\|\cdot\|_{L^2(U)}$, then the sequence $\{f_n\}$ converges uniformly on every compact subset of U to a holomorphic function.

[Hint: Use the mean-value property.]

21. Certain sets have geometric properties that guarantee they are simply connected.

(a) An open set $\Omega \subset \mathbb{C}$ is **convex** if for any two points in Ω, the straight line segment between them is contained in Ω. Prove that a convex open set is simply connected.

(b) More generally, an open set $\Omega \subset \mathbb{C}$ is **star-shaped** if there exists a point $z_0 \in \Omega$ such that for any $z \in \Omega$, the straight line segment between z and z_0 is contained in Ω. Prove that a star-shaped open set is simply connected. Conclude that the slit plane $\mathbb{C} - \{(-\infty, 0]\}$ (and more generally any sector, convex or not) is simply connected.

(c) What are other examples of open sets that are simply connected?

22. Show that there is no holomorphic function f in the unit disc $\mathbb{D}$ that extends continuously to $\partial\mathbb{D}$ such that $f(z) = 1/z$ for $z \in \partial\mathbb{D}$.

9 Problems

1.* Consider a holomorphic map on the unit disc $f : \mathbb{D} \to \mathbb{C}$ which satisfies $f(0) = 0$. By the open mapping theorem, the image $f(\mathbb{D})$ contains a small disc centered at the origin. We then ask: does there exist $r > 0$ such that for *all* $f : \mathbb{D} \to \mathbb{C}$ with $f(0) = 0$, we have $D_r(0) \subset f(\mathbb{D})$?

(a) Show that with no further restrictions on f, no such r exists. It suffices to find a sequence of functions $\{f_n\}$ holomorphic in $\mathbb{D}$ such that $1/n \notin f(\mathbb{D})$. Compute $f_n'(0)$, and discuss.

(b) Assume in addition that f also satisfies $f'(0) = 1$. Show that despite this new assumption, there exists no $r > 0$ satisfying the desired condition.

[Hint: Try $f_\epsilon(z) = \epsilon(e^{z/\epsilon} - 1)$.]

The Koebe-Bieberbach theorem states that if in addition to $f(0) = 0$ and $f'(0) = 1$ we also assume that f is injective, then such an r exists and the best possible value is $r = 1/4$.

(c) As a first step, show that if $h(z) = \frac{1}{z} + c_0 + c_1 z + c_2 z^2 + \cdots$ is analytic and injective for $0 < |z| < 1$, then $\sum_{n=1}^{\infty} n|c_n|^2 \leq 1$.

[Hint: Calculate the area of the complement of $h(D_\rho(0) - \{0\})$ where $0 < \rho < 1$, and let $\rho \to 1$.]

(d) If $f(z) = z + a_2 z^2 + \cdots$ satisfies the hypotheses of the theorem, show that there exists another function g satisfying the hypotheses of the theorem such that $g^2(z) = f(z^2)$.

[Hint: $f(z)/z$ is nowhere vanishing so there exists ψ such that $\psi^2(z) = f(z)/z$ and $\psi(0) = 1$. Check that $g(z) = z\psi(z^2)$ is injective.]

(e) With the notation of the previous part, show that $|a_2| \leq 2$, and that equality holds if and only if

$$f(z) = \frac{z}{(1 - e^{i\theta} z)^2} \qquad \text{for some } \theta \in \mathbb{R}.$$

[Hint: What is the power series expansion of $1/g(z)$? Use part (c).]

(f) If $h(z) = \frac{1}{z} + c_0 + c_1 z + c_2 z^2 + \cdots$ is injective on $\mathbb{D}$ and avoids the values z_1 and z_2, show that $|z_1 - z_2| \leq 4$.

[Hint: Look at the second coefficient in the power series expansion of $1/(h(z) - z_j)$.]

(g) Complete the proof of the theorem. [Hint: If f avoids w, then $1/f$ avoids 0 and $1/w$.]

2. Let u be a harmonic function in the unit disc that is continuous on its closure. Deduce Poisson's integral formula

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - |z_0|^2}{|e^{i\theta} - z_0|^2} u(e^{i\theta})\, d\theta \qquad \text{for } |z_0| < 1$$

from the special case $z_0 = 0$ (the mean value theorem). Show that if $z_0 = re^{i\varphi}$, then

$$\frac{1 - |z_0|^2}{|e^{i\theta} - z_0|^2} = \frac{1 - r^2}{1 - 2r\cos(\theta - \varphi) + r^2} = P_r(\theta - \varphi),$$

and we recover the expression for the Poisson kernel derived in the exercises of the previous chapter.

[Hint: Set $u_0(z) = u(T(z))$ where

$$T(z) = \frac{z_0 - z}{1 - \overline{z_0} z}.$$

Prove that u_0 is harmonic. Then apply the mean value theorem to u_0, and make a change of variables in the integral.]

3. If $f(z)$ is holomorphic in the deleted neighborhood $\{0 < |z - z_0| < r\}$ and has a pole of order k at z_0, then we can write

$$f(z) = \frac{a_{-k}}{(z - z_0)^k} + \cdots + \frac{a_{-1}}{(z - z_0)} + g(z)$$

where g is holomorphic in the disc $\{|z - z_0| < r\}$. There is a generalization of this expansion that holds even if z_0 is an essential singularity. This is a special case of the **Laurent series expansion**, which is valid in an even more general setting.

Let f be holomorphic in a region containing the annulus $\{z : r_1 \leq |z - z_0| \leq r_2\}$ where $0 < r_1 < r_2$. Then,

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

where the series converges absolutely in the interior of the annulus. To prove this, it suffices to write

$$f(z) = \frac{1}{2\pi i} \int_{C_{r_2}} \frac{f(\zeta)}{\zeta - z}\, d\zeta - \frac{1}{2\pi i} \int_{C_{r_1}} \frac{f(\zeta)}{\zeta - z}\, d\zeta$$

when $r_1 < |z - z_0| < r_2$, and argue as in the proof of Theorem 4.4, Chapter 2. Here C_{r_1} and C_{r_2} are the circles bounding the annulus.

4.* Suppose Ω is a bounded region. Let L be a (two-way infinite) line that intersects Ω. Assume that $\Omega \cap L$ is an interval I. Choosing an orientation for L, we can define Ω_l and Ω_r to be the subregions of Ω lying strictly to the left or right of L, with $\Omega = \Omega_l \cup I \cup \Omega_r$ a disjoint union. If Ω_l and Ω_r are simply connected, then Ω is simply connected.

5.* Let

$$g(z) = \frac{1}{2\pi i} \int_{-M}^{M} \frac{h(x)}{x-z}\, dx$$

where h is continuous and supported in $[-M, M]$.

(a) Prove that the function g is holomorphic in $\mathbb{C} - [-M, M]$, and vanishes at infinity, that is, $\lim_{|z|\to\infty} |g(z)| = 0$. Moreover, the "jump" of g across $[-M, M]$ is h, that is,

$$h(x) = \lim_{\epsilon\to 0, \epsilon>0} g(x + i\epsilon) - g(x - i\epsilon).$$

[Hint: Express the difference $g(x + i\epsilon) - g(x - i\epsilon)$ in terms of a convolution of h with the Poisson kernel.]

(b) If h satisfies a mild smoothness condition, for instance a Hölder condition with exponent α, that is, $|h(x) - h(y)| \leq C|x - y|^{\alpha}$ for some $C > 0$ and all $x, y \in [-M, M]$, then $g(x + i\epsilon)$ and $g(x - i\epsilon)$ converge uniformly to functions $g_+(x)$ and $g_-(x)$ as $\epsilon \to 0$. Then, g can be characterized as the unique holomorphic function that satisfies:

(i) g is holomorphic outside $[-M, M]$,

(ii) g vanishes at infinity,

(iii) $g(x + i\epsilon)$ and $g(x - i\epsilon)$ converge uniformly as $\epsilon \to 0$ to functions $g_+(x)$ and $g_-(x)$ with

$$g_+(x) - g_-(x) = h(x).$$

[Hint: If G is another function satisfying these conditions, $g - G$ is entire.]

4 The Fourier Transform

Raymond Edward Alan Christopher Paley, Fellow of Trinity College, Cambridge, and International Research Fellow at the Massachusetts Institute of Technology and at Harvard University, was killed by an avalanche on April 7, 1933, while skiing in the vicinity of Banff, Alberta. Although only twenty-six years of age, he was already recognized as the ablest of the group of young English mathematicians who have been inspired by the genius of G. H. Hardy and J. E. Littlewood. In a group notable for its brilliant technique, no one had developed this technique to a higher degree than Paley. Nevertheless he should not be thought of primarily as a technician, for with this ability he combined creative power of the first order. As he himself was wont to say, technique without "rugger tactics" will not get one far, and these rugger tactics he practiced to a degree that was characteristic of his forthright and vigorous nature.

Possessed of an extraordinary capacity for making friends and for scientific collaboration, Paley believed that the inspiration of continual interchange of ideas stimulates each collaborator to accomplish more than he would alone. Only the exceptional man works well with a partner, but Paley had collaborated successfully with many, including Littlewood, Pólya, Zygmund, and Wiener.

N. Wiener, 1933

If f is a function on $\mathbb{R}$ that satisfies appropriate regularity and decay conditions, then its Fourier transform is defined by

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x\xi}\, dx, \qquad \xi \in \mathbb{R}$$

and its counterpart, the Fourier inversion formula, holds

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi, \qquad x \in \mathbb{R}.$$

The Fourier transform (including its d-dimensional variants), plays a basic role in analysis, as the reader of Book I is aware. Here we want to illustrate the intimate and fruitful connection between the one-dimensional theory of the Fourier transform and complex analysis. The main theme (stated somewhat imprecisely) is as follows: for a function f initially defined on the real line, the possibility of extending it to a holomorphic function is closely related to the very rapid (for example, exponential) decay at infinity of its Fourier transform $\hat{f}$. We elaborate on this theme in two stages.

First, we assume that f can be analytically continued in a horizontal strip containing the real axis, and has "moderate decrease" at infinity,[1] so that the integral defining the Fourier transform $\hat{f}$ converges. As a result, we conclude that $\hat{f}$ decreases exponentially at infinity; it also follows directly that the Fourier inversion formula holds. Moreover one can easily obtain from these considerations the Poisson summation formula $\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \hat{f}(n)$. Incidentally, all these theorems are elegant consequences of contour integration.

At a second stage, we take as our starting point the validity of the Fourier inversion formula, which holds if we assume that both f and $\hat{f}$ are of moderate decrease, without making any assumptions on the analyticity of f. We then ask a simple but natural question: What are the conditions on f so that its Fourier transform is supported in a bounded interval, say $[-M, M]$? This is a basic problem that, as one notices, can be stated without any reference to notions of complex analysis. However, it can be resolved only in terms of the holomorphic properties of the function f. The condition, given by the Paley-Wiener theorem, is that there be a holomorphic extension of f to $\mathbb{C}$ that satisfies the growth condition

$$|f(z)| \leq Ae^{2\pi M|z|} \quad \text{for some constant } A > 0.$$

Functions satisfying this condition are said to be of **exponential type**.

Observe that the condition that $\hat{f}$ vanishes outside a compact set can be viewed as an extreme version of a decay property at infinity, and so the above theorem clearly falls within the context of the theme indicated above.

In all these matters a decisive technique will consist in shifting the contour of integration, that is the real line, within the boundaries of a horizontal strip. This will take advantage of the special behavior of $e^{-2\pi i z\xi}$ when z has a non-zero imaginary part. Indeed, when z is real this exponential remains bounded and oscillates, while if $\mathrm{Im}(z) \neq 0$, it will

[1]We say that a function f is of **moderate decrease** if f is continuous and there exists $A > 0$ so that $|f(x)| \leq A/(1 + x^2)$ for all $x \in \mathbb{R}$. A more restrictive condition is that $f \in \mathcal{S}$, the Schwartz space of testing functions, which also implies that $\hat{f}$ belongs to $\mathcal{S}$. See Book I for more details.

have exponential decay or exponential increase, depending on whether the product $\xi\text{Im}(z)$ is negative or positive.

1 The class $\mathfrak{F}$

The weakest decay condition imposed on functions in our study of the Fourier transform in Book I was that of moderate decrease. There, we proved the Fourier inversion and Poisson summation formulas under the hypothesis that f and $\hat{f}$ satisfy

$$|f(x)| \leq \frac{A}{1+x^2} \quad \text{and} \quad |\hat{f}(\xi)| \leq \frac{A'}{1+\xi^2}$$

for some positive constants A, A' and all $x, \xi \in \mathbb{R}$. We were led to consider this class of functions because of various examples such as the Poisson kernel

$$P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$$

for $y > 0$, which played a fundamental role in the solution of the Dirichlet problem for the steady-state heat equation in the upper half-plane. There we had $\widehat{P_y}(\xi) = e^{-2\pi y|\xi|}$.

In the present context, we introduce a class of functions particularly suited to the goal we have set: proving theorems about the Fourier transform using complex analysis. Moreover, this class will be large enough to contain many of the important applications we have in mind.

For each $a > 0$ we denote by $\mathfrak{F}_a$ the class of all functions f that satisfy the following two conditions:

(i) The function f is holomorphic in the horizontal strip

$$S_a = \{z \in \mathbb{C} : \ |\text{Im}(z)| < a\}.$$

(ii) There exists a constant $A > 0$ such that

$$|f(x+iy)| \leq \frac{A}{1+x^2} \quad \text{for all } x \in \mathbb{R} \text{ and } |y| < a.$$

In other words, $\mathfrak{F}_a$ consists of those holomorphic functions on S_a that are of moderate decay on each horizontal line $\text{Im}(z) = y$, uniformly in $-a < y < a$. For example, $f(z) = e^{-\pi z^2}$ belongs to $\mathfrak{F}_a$ for all a. Also, the function

$$f(z) = \frac{1}{\pi}\frac{c}{c^2+z^2},$$

which has simple poles at $z = \pm ci$, belongs to $\mathfrak{F}_a$ for all $0 < a < c$.

Another example is provided by $f(z) = 1/\cosh \pi z$, which belongs to $\mathfrak{F}_a$ whenever $|a| < 1/2$. This function, as well as one of its fundamental properties, was already discussed in Example 3, Section 2.1 of Chapter 3.

Note also that a simple application of the Cauchy integral formulas shows that if $f \in \mathfrak{F}_a$, then for every n, the n^{th} derivative of f belongs to $\mathfrak{F}_b$ for all b with $0 < b < a$ (Exercise 2).

Finally, we denote by $\mathfrak{F}$ the class of all functions that belong to $\mathfrak{F}_a$ for some a.

Remark. The condition of moderate decrease can be weakened somewhat by replacing the order of decrease of $A/(1+x^2)$ by $A/(1+|x|^{1+\epsilon})$ for any $\epsilon > 0$. As the reader will observe, many of the results below remain unchanged with this less restrictive condition.

2 Action of the Fourier transform on $\mathfrak{F}$

Here we prove three theorems, including the Fourier inversion and Poisson summation formulas, for functions in $\mathfrak{F}$. The idea behind all three proofs is the same: contour integration. Thus the approach used will be different from that of the corresponding results in Book I.

Theorem 2.1 *If f belongs to the class $\mathfrak{F}_a$ for some $a > 0$, then $|\hat{f}(\xi)| \leq Be^{-2\pi b|\xi|}$ for any $0 \leq b < a$.*

Proof. Recall that $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i x\xi}\, dx$. The case $b = 0$ simply says that $\hat{f}$ is bounded, which follows at once from the integral defining $\hat{f}$, the assumption that f is of moderate decrease, and the fact that the exponential is bounded by 1.

Now suppose $0 < b < a$ and assume first that $\xi > 0$. The main step consists of shifting the contour of integration, that is the real line, down by b. More precisely, consider the contour in Figure 1 as well as the function $g(z) = f(z)e^{-2\pi i z\xi}$.

We claim that as R tends to infinity, the integrals of g over the two vertical sides converge to zero. For example, the integral over the vertical segment on the left can be estimated by

$$\begin{aligned}\left|\int_{-R-ib}^{-R} g(z)\, dz\right| &\leq \int_0^b |f(-R-it)e^{-2\pi i(-R-it)\xi}|\, dt \\ &\leq \int_0^b \frac{A}{R^2} e^{-2\pi t\xi}\, dt \\ &= O(1/R^2).\end{aligned}$$

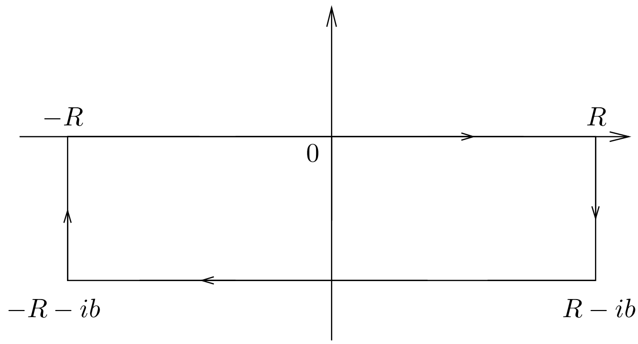

Figure 1. The contour in the proof of Theorem 2.1 when $\xi > 0$

A similar estimate for the other side proves our claim. Therefore, by Cauchy's theorem applied to the large rectangle, we find in the limit as R tends to infinity that

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x-ib)e^{-2\pi i(x-ib)\xi}\,dx, \tag{1}$$

which leads to the estimate

$$|\hat{f}(\xi)| \leq \int_{-\infty}^{\infty} \frac{A}{1+x^2} e^{-2\pi b\xi}\,dx \leq Be^{-2\pi b\xi},$$

where B is a suitable constant. A similar argument for $\xi < 0$, but this time shifting the real line up by b, allows us to finish the proof of the theorem.

This result says that whenever $f \in \mathfrak{F}$, then $\hat{f}$ has rapid decay at infinity. We remark that the further we can extend f (that is, the larger a), then the larger we can choose b, hence the better the decay. We shall return to this circle of ideas in Section 3, where we describe those f for which $\hat{f}$ has the ultimate decay condition: compact support.

Since $\hat{f}$ decreases rapidly on $\mathbb{R}$, the integral in the Fourier inversion formula makes sense, and we now turn to the complex analytic proof of this identity.

Theorem 2.2 *If $f \in \mathfrak{F}$, then the Fourier inversion holds, namely*

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi i x\xi}\,d\xi \qquad \textit{for all } x \in \mathbb{R}.$$

Besides contour integration, the proof of the theorem requires a simple identity, which we isolate.

Lemma 2.3 *If A is positive and B is real, then* $\int_0^\infty e^{-(A+iB)\xi}\,d\xi = \frac{1}{A+iB}$.

Proof. Since $A > 0$ and $B \in \mathbb{R}$, we have $|e^{-(A+iB)\xi}| = e^{-A\xi}$, and the integral converges. By definition

$$\int_0^\infty e^{-(A+iB)\xi}\,d\xi = \lim_{R\to\infty}\int_0^R e^{-(A+iB)\xi}\,d\xi.$$

However,

$$\int_0^R e^{-(A+iB)\xi}\,d\xi = \left[-\frac{e^{-(A+iB)\xi}}{A+iB}\right]_0^R,$$

which tends to $1/(A+iB)$ as R tends to infinity.

We can now prove the inversion theorem. Once again, the sign of ξ matters, so we begin by writing

$$\int_{-\infty}^\infty \hat{f}(\xi)e^{2\pi ix\xi}\,d\xi = \int_{-\infty}^0 \hat{f}(\xi)e^{2\pi ix\xi}\,d\xi + \int_0^\infty \hat{f}(\xi)e^{2\pi ix\xi}\,d\xi.$$

For the second integral we argue as follows. Say $f \in \mathfrak{F}_a$ and choose $0 < b < a$. Arguing as the proof of Theorem 2.1, or simply using equation (1), we get

$$\hat{f}(\xi) = \int_{-\infty}^\infty f(u-ib)e^{-2\pi i(u-ib)\xi}\,du,$$

so that with an application of the lemma and the convergence of the integration in ξ, we find

$$\begin{aligned}
\int_0^\infty \hat{f}(\xi)e^{2\pi ix\xi}\,d\xi &= \int_0^\infty\int_{-\infty}^\infty f(u-ib)e^{-2\pi i(u-ib)\xi}e^{2\pi ix\xi}\,du\,d\xi\\
&= \int_{-\infty}^\infty f(u-ib)\int_0^\infty e^{-2\pi i(u-ib-x)\xi}\,d\xi\,du\\
&= \int_{-\infty}^\infty f(u-ib)\frac{1}{2\pi b + 2\pi i(u-x)}\,du\\
&= \frac{1}{2\pi i}\int_{-\infty}^\infty \frac{f(u-ib)}{u-ib-x}\,du\\
&= \frac{1}{2\pi i}\int_{L_1}\frac{f(\zeta)}{\zeta - x}\,d\zeta,
\end{aligned}$$

where L_1 denotes the line $\{u - ib : u \in \mathbb{R}\}$ traversed from left to right. (In other words, L_1 is the real line shifted down by b.) For the integral when $\xi < 0$, a similar calculation gives

$$\int_{-\infty}^{0} \hat{f}(\xi) e^{2\pi i x \xi}\, d\xi = -\frac{1}{2\pi i} \int_{L_2} \frac{f(\zeta)}{\zeta - x}\, d\zeta,$$

where L_2 is the real line shifted up by b, with orientation from left to right. Now given $x \in \mathbb{R}$, consider the contour γ_R in Figure 2.

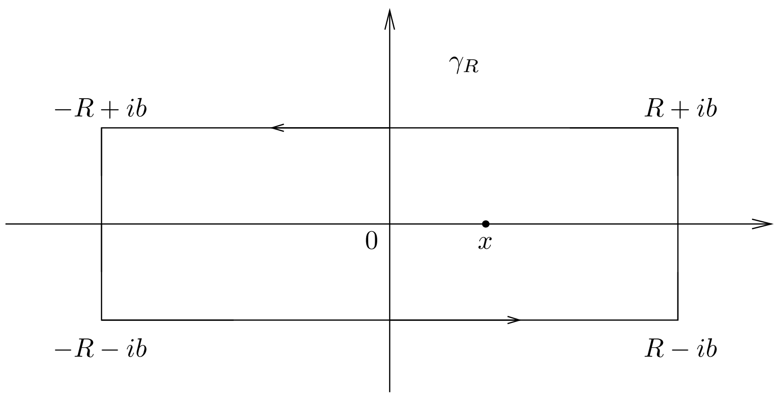

Figure 2. The contour γ_R in the proof of Theorem 2.2

The function $f(\zeta)/(\zeta - x)$ has a simple pole at x with residue $f(x)$, so the residue formula gives

$$f(x) = \frac{1}{2\pi i} \int_{\gamma_R} \frac{f(\zeta)}{\zeta - x}\, d\zeta.$$

Letting R tend to infinity, one checks easily that the integral over the vertical sides goes to 0 and therefore, combining with the previous results, we get

$$\begin{aligned} f(x) &= \frac{1}{2\pi i} \int_{L_1} \frac{f(\zeta)}{\zeta - x}\, d\zeta - \frac{1}{2\pi i} \int_{L_2} \frac{f(\zeta)}{\zeta - x}\, d\zeta \\ &= \int_{0}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi}\, d\xi + \int_{-\infty}^{0} \hat{f}(\xi) e^{2\pi i x \xi}\, d\xi \\ &= \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi}\, d\xi, \end{aligned}$$

and the theorem is proved.

The last of our three theorems is the Poisson summation formula.

Theorem 2.4 *If $f \in \mathfrak{F}$, then*

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \hat{f}(n).$$

Proof. Say $f \in \mathfrak{F}_a$ and choose some b satisfying $0 < b < a$. The function $1/(e^{2\pi iz} - 1)$ has simple poles with residue $1/(2\pi i)$ at the integers. Thus $f(z)/(e^{2\pi iz} - 1)$ has simple poles at the integers n, with residues $f(n)/2\pi i$. We may therefore apply the residue formula to the contour γ_N in Figure 3 where N is an integer.

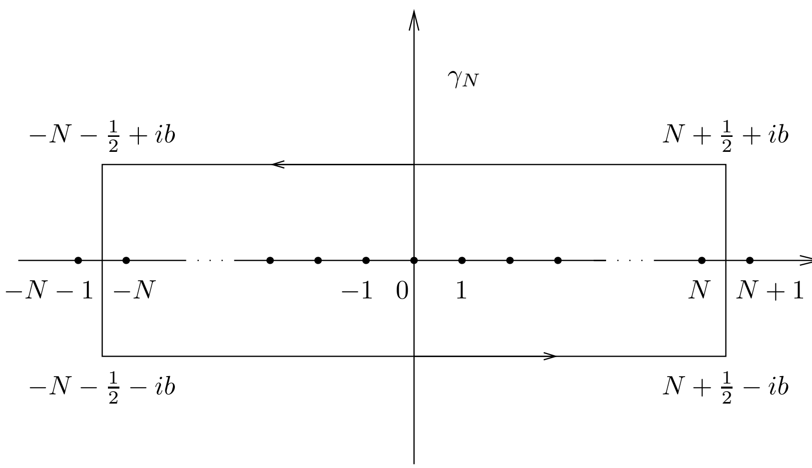

Figure 3. The contour γ_N in the proof of Theorem 2.4

This yields

$$\sum_{|n|\leq N} f(n) = \int_{\gamma_N} \frac{f(z)}{e^{2\pi iz} - 1}\, dz.$$

Letting N tend to infinity, and recalling that f has moderate decrease, we see that the sum converges to $\sum_{n\in\mathbb{Z}} f(n)$, and also that the integral over the vertical segments goes to 0. Therefore, in the limit we get

$$\sum_{n\in\mathbb{Z}} f(n) = \int_{L_1} \frac{f(z)}{e^{2\pi iz} - 1}\, dz - \int_{L_2} \frac{f(z)}{e^{2\pi iz} - 1}\, dz \tag{2}$$

where L_1 and L_2 are the real line shifted down and up by b, respectively.

Now we use the fact that if $|w| > 1$, then

$$\frac{1}{w-1} = w^{-1}\sum_{n=0}^{\infty} w^{-n}$$

to see that on L_1 (where $|e^{2\pi iz}| > 1$) we have

$$\frac{1}{e^{2\pi iz}-1} = e^{-2\pi iz}\sum_{n=0}^{\infty} e^{-2\pi inz}.$$

Also if $|w| < 1$, then

$$\frac{1}{w-1} = -\sum_{n=0}^{\infty} w^{n}$$

so that on L_2

$$\frac{1}{e^{2\pi iz}-1} = -\sum_{n=0}^{\infty} e^{2\pi inz}.$$

Substituting these observations in (2) we find that

$$\begin{aligned}
\sum_{n\in\mathbb{Z}} f(n) &= \int_{L_1} f(z)\left(e^{-2\pi iz}\sum_{n=0}^{\infty} e^{-2\pi inz}\right)dz + \int_{L_2} f(z)\left(\sum_{n=0}^{\infty} e^{2\pi inz}\right)dz \\
&= \sum_{n=0}^{\infty}\int_{L_1} f(z)e^{-2\pi i(n+1)z}\,dz + \sum_{n=0}^{\infty}\int_{L_2} f(z)e^{2\pi inz}\,dz \\
&= \sum_{n=0}^{\infty}\int_{-\infty}^{\infty} f(x)e^{-2\pi i(n+1)x}\,dx + \sum_{n=0}^{\infty}\int_{-\infty}^{\infty} f(x)e^{2\pi inx}\,dz \\
&= \sum_{n=0}^{\infty}\hat{f}(n+1) + \sum_{n=0}^{\infty}\hat{f}(-n) \\
&= \sum_{n\in\mathbb{Z}}\hat{f}(n),
\end{aligned}$$

where we have shifted L_1 and L_2 back to the real line according to equation (1) and its analogue for the shift down.

The Poisson summation formula has many far-reaching consequences, and we close this section by deriving several interesting identities that are of importance for later applications.

First, we recall the calculation in Example 1, Chapter 2, which showed that the function $e^{-\pi x^2}$ was its own Fourier transform:

$$\int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx = e^{-\pi\xi^2}.$$

For fixed values of $t > 0$ and $a \in \mathbb{R}$, the change of variables $x \mapsto t^{1/2}(x + a)$ in the above integral shows that the Fourier transform of the function $f(x) = e^{-\pi t(x+a)^2}$ is $\hat{f}(\xi) = t^{-1/2}e^{-\pi\xi^2/t}e^{2\pi i a\xi}$. Applying the Poisson summation formula to the pair f and $\hat{f}$ (which belong to $\mathfrak{F}$) provides the following relation:

$$\sum_{n=-\infty}^{\infty} e^{-\pi t(n+a)^2} = \sum_{n=-\infty}^{\infty} t^{-1/2}e^{-\pi n^2/t}e^{2\pi i n a}. \tag{3}$$

This identity has noteworthy consequences. For instance, the special case $a = 0$ is the transformation law for a version of the "theta function": if we define ϑ for $t > 0$ by the series $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$, then the relation (3) says precisely that

$$\vartheta(t) = t^{-1/2}\vartheta(1/t) \qquad \text{for } t > 0. \tag{4}$$

This equation will be used in Chapter 6 to derive the key functional equation of the Riemann zeta function, and this leads to its analytic continuation. Also, the general case $a \in \mathbb{R}$ will be used in Chapter 10 to establish a corresponding law for the more general Jacobi theta function Θ.

For another application of the Poisson summation formula we recall that we proved in Example 3, Chapter 3, that the function $1/\cosh \pi x$ was also its own Fourier transform:

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x\xi}}{\cosh \pi x}\, dx = \frac{1}{\cosh \pi\xi}.$$

This implies that if $t > 0$ and $a \in \mathbb{R}$, then the Fourier transform of the function $f(x) = e^{-2\pi i a x}/\cosh(\pi x/t)$ is $\hat{f}(\xi) = t/\cosh(\pi(\xi + a)t)$, and the Poisson summation formula yields

$$\sum_{n=-\infty}^{\infty} \frac{e^{-2\pi i a n}}{\cosh(\pi n/t)} = \sum_{n=-\infty}^{\infty} \frac{t}{\cosh(\pi(n + a)t)}. \tag{5}$$

This formula will be used in Chapter 10 in the context of the two-squares theorem.

3 Paley-Wiener theorem

In this section we change our point of view somewhat: we do not suppose any analyticity of f, but we do assume the validity of the Fourier inversion formula

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi \quad \text{if} \quad \hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x\xi}\, dx,$$

under the conditions $|f(x)| \leq A/(1+x^2)$ and $|\hat{f}(\xi)| \leq A'/(1+\xi^2)$. For a proof of the inversion formula under these conditions, we refer the reader to Chapter 5 in Book I.

We start by pointing out a partial converse to Theorem 2.1.

Theorem 3.1 *Suppose $\hat{f}$ satisfies the decay condition $|\hat{f}(\xi)| \leq Ae^{-2\pi a|\xi|}$ for some constants $a, A > 0$. Then $f(x)$ is the restriction to $\mathbb{R}$ of a function $f(z)$ holomorphic in the strip $S_b = \{z \in \mathbb{C} : \ |\mathrm{Im}(z)| < b\}$, for any $0 < b < a$.*

Proof. Define

$$f_n(z) = \int_{-n}^{n} \hat{f}(\xi) e^{2\pi i \xi z}\, d\xi,$$

and note that f_n is entire by Theorem 5.4 in Chapter 2. Observe also that $f(z)$ may be defined for all z in the strip S_b by

$$f(z) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi z}\, d\xi\,,$$

because the integral converges absolutely by our assumption on $\hat{f}$: it is majorized by

$$A \int_{-\infty}^{\infty} e^{-2\pi a|\xi|} e^{2\pi b|\xi|}\, d\xi\,,$$

which is finite if $b < a$. Moreover, for $z \in S_b$

$$\begin{aligned} |f(z) - f_n(z)| &\leq A \int_{|\xi| \geq n} e^{-2\pi a|\xi|} e^{2\pi b|\xi|}\, d\xi \\ &\to 0 \quad \text{as } n \to \infty, \end{aligned}$$

and thus the sequence $\{f_n\}$ converges to f uniformly in S_b, which, by Theorem 5.2 in Chapter 2, proves the theorem.

We digress briefly to make the following observation.

Corollary 3.2 *If $\hat{f}(\xi) = O(e^{-2\pi a|\xi|})$ for some $a > 0$, and f vanishes in a non-empty open interval, then $f = 0$.*

Since by the theorem f is analytic in a region containing the real line, the corollary is a consequence of Theorem 4.8 in Chapter 2. In particular, we recover the fact proved in Exercise 21, Chapter 5 in Book I, namely that f and $\hat{f}$ cannot both have compact support unless $f = 0$.

The Paley-Wiener theorem goes a step further than the previous theorem, and describes the nature of those functions whose Fourier transforms are supported in a given interval $[-M, M]$.

Theorem 3.3 *Suppose f is continuous and of moderate decrease on $\mathbb{R}$. Then, f has an extension to the complex plane that is entire with $|f(z)| \leq Ae^{2\pi M|z|}$ for some $A > 0$, if and only if $\hat{f}$ is supported in the interval $[-M, M]$.*

One direction is simple. Suppose $\hat{f}$ is supported in $[-M, M]$. Then both f and $\hat{f}$ have moderate decrease, and the Fourier inversion formula applies

$$f(x) = \int_{-M}^{M} \hat{f}(\xi)e^{2\pi i\xi x}\, d\xi.$$

Since the range of integration is finite, we may replace x by the complex variable z in the integral, thereby defining a complex-valued function on $\mathbb{C}$ by

$$g(z) = \int_{-M}^{M} \hat{f}(\xi)e^{2\pi i\xi z}\, d\xi.$$

By construction $g(z) = f(z)$ if z is real, and g is holomorphic by Theorem 5.4 in Chapter 2. Finally, if $z = x + iy$, we have

$$\begin{aligned} |g(z)| &\leq \int_{-M}^{M} |\hat{f}(\xi)|e^{-2\pi\xi y}\, d\xi \\ &\leq Ae^{2\pi M|z|}. \end{aligned}$$

The converse result requires a little more work. It starts with the observation that if $\hat{f}$ were supported in $[-M, M]$, then the argument above would give the stronger bound $|f(z)| \leq Ae^{2\pi|y|}$ instead of what we assume, that is $|f(z)| \leq Ae^{2\pi|z|}$. The idea is then to try to reduce to the better situation, where this stronger bound holds. However, this is not quite enough because we need in addition a (moderate) decay as $x \to \infty$

(when $y \neq 0$) to deal with the convergence of certain integrals at infinity. Thus we begin by also assuming this further property of f, and then we remove the additional assumptions, one step at a time.

Step 1. We first assume that f is holomorphic in the complex plane, and satisfies the following condition regarding decay in x and growth in y:

$$|f(x+iy)| \leq A' \frac{e^{2\pi M|y|}}{1+x^2}. \tag{6}$$

We then prove under this stronger assumption that $\hat{f}(\xi) = 0$ if $|\xi| > M$. To see this, we first suppose that $\xi > M$ and write

$$\begin{aligned} \hat{f}(\xi) &= \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x}\, dx \\ &= \int_{-\infty}^{\infty} f(x-iy) e^{-2\pi i \xi (x-iy)}\, dx. \end{aligned}$$

Here we have shifted the real line down by an amount $y > 0$ using the standard argument (equation (1)). Putting absolute values gives the bound

$$\begin{aligned} |\hat{f}(\xi)| &\leq A' \int_{-\infty}^{\infty} \frac{e^{2\pi M y - 2\pi \xi y}}{1+x^2}\, dx \\ &\leq C e^{-2\pi y(\xi - M)}. \end{aligned}$$

Letting y tend to infinity, and recalling that $\xi - M > 0$, proves that $\hat{f}(\xi) = 0$. A similar argument, shifting the contour up by $y > 0$, proves that $\hat{f}(\xi) = 0$ whenever $\xi < -M$.

Step 2. We relax condition (6) by assuming only that f satisfies

$$|f(x+iy)| \leq A e^{2\pi M|y|}. \tag{7}$$

This is still a stronger condition than in the theorem, but it is weaker than (6). Suppose first that $\xi > M$, and for $\epsilon > 0$ consider the following auxiliary function

$$f_\epsilon(z) = \frac{f(z)}{(1+i\epsilon z)^2}.$$

We observe that the quantity $1/(1+i\epsilon z)^2$ has absolute value less than or equal to 1 in the closed lower half-plane (including the real line) and converges to 1 as ϵ tends to 0. In particular, this shows that $\widehat{f_\epsilon}(\xi) \to \hat{f}(\xi)$ as $\epsilon \to 0$ since we may write

$$|\widehat{f_\epsilon}(\xi) - \hat{f}(\xi)| \leq \int_{-\infty}^{\infty} |f(x)| \left[\frac{1}{(1+i\epsilon x)^2} - 1\right] dx,$$

and recall that f has moderate decrease on $\mathbb{R}$.

But for each fixed ϵ, we have

$$|f_\epsilon(x+iy)| \leq A'' \frac{e^{2\pi M|y|}}{1+x^2},$$

so by Step 1 we must have $\widehat{f_\epsilon}(\xi) = 0$, and hence $\hat{f}(\xi) = 0$ after passing to the limit as $\epsilon \to 0$. A similar argument applies if $\xi < -M$, although we must now argue in the upper half-plane, and use the factor $1/(1 - i\epsilon z)^2$ instead.

Step 3. To conclude the proof, it suffices to show that the conditions in the theorem imply condition (7) in Step 2. In fact, after dividing by an appropriate constant, it suffices to show that if $|f(x)| \leq 1$ for all real x, and $|f(z)| \leq e^{2\pi M|z|}$ for all complex z, then

$$|f(x+iy)| \leq e^{2\pi M|y|}.$$

This will follow from an ingenious and very useful idea of Phragmén and Lindelöf that allows one to adapt the maximum modulus principle to various unbounded regions. The particular result we need is as follows.

Theorem 3.4 *Suppose F is a holomorphic function in the sector*

$$S = \{z : -\pi/4 < \arg z < \pi/4\}$$

that is continuous on the closure of S. Assume $|F(z)| \leq 1$ on the boundary of the sector, and that there are constants $C, c > 0$ such that $|F(z)| \leq Ce^{c|z|}$ for all z in the sector. Then

$$|F(z)| \leq 1 \quad \textit{for all } z \in S.$$

In other words, if F is bounded by 1 on the boundary of S and has no more than a reasonable amount of growth, then F is actually bounded everywhere by 1. That some restriction on the growth of F is necessary follows from a simple observation. Consider the function $F(z) = e^{z^2}$. Then F is bounded by 1 on the boundary of S, but if x is real, $F(x)$ is unbounded as $x \to \infty$. We now give the proof of Theorem 3.4.

Proof. The idea is to subdue the "enemy" function e^{z^2} and turn it to our advantage: in brief, one modifies e^{z^2} by replacing it by e^{z^α} with $\alpha < 2$. For simplicity we take the case $\alpha = 3/2$.

If $\epsilon > 0$, let

$$F_\epsilon(z) = F(z)e^{-\epsilon z^{3/2}}.$$

Here we have chosen the principal branch of the logarithm to define $z^{3/2}$ so that if $z = re^{i\theta}$ (with $-\pi < \theta < \pi$), then $z^{3/2} = r^{3/2}e^{3i\theta/2}$. Hence F_ϵ is holomorphic in S and continuous up to its boundary. Also

$$|e^{-\epsilon z^{3/2}}| = e^{-\epsilon r^{3/2}\cos(3\theta/2)};$$

and since $-\pi/4 < \theta < \pi/4$ in the sector, we get the inequalities

$$-\frac{\pi}{2} < -\frac{3\pi}{8} < \frac{3\theta}{2} < \frac{3\pi}{8} < \frac{\pi}{2},$$

and therefore $\cos(3\theta/2)$ is strictly positive in the sector. This, together with the fact that $|F(z)| \leq Ce^{c|z|}$, shows that $F_\epsilon(z)$ decreases rapidly in the closed sector as $|z| \to \infty$, and in particular F_ϵ is bounded. We claim that in fact $|F_\epsilon(z)| \leq 1$ for all $z \in \overline{S}$, where $\overline{S}$ denotes the closure of S. To prove this, we define

$$M = \sup_{z \in \overline{S}} |F_\epsilon(z)|.$$

Assuming F is not identically zero, let $\{w_j\}$ be a sequence of points such that $|F_\epsilon(w_j)| \to M$. Since $M \neq 0$ and F_ϵ decays to 0 as $|z|$ becomes large in the sector, w_j cannot escape to infinity, and we conclude that this sequence accumulates to a point $w \in \overline{S}$. By the maximum principle, w cannot be an interior point of S, so w lies on its boundary. But on the boundary, we have first $|F(z)| \leq 1$ by assumption, and also $|e^{-\epsilon z^{3/2}}| \leq 1$, so that $M \leq 1$, and the claim is proved.

Finally, we may let ϵ tend to 0 to conclude the proof of the theorem.

Further generalizations of the Phragmén-Lindelöf theorem are included in Exercise 9 and Problem 3.

We must now use this result to conclude the proof of the Paley-Wiener theorem, that is, show that if $|f(x)| \leq 1$ and $|f(z)| \leq e^{2\pi M|z|}$, then $|f(z)| \leq e^{2\pi M|y|}$. First, note that the sector in the Phragmén-Lindelöf theorem can be rotated, say to the first quadrant $Q = \{z = x + iy :\ x > 0,\ y > 0\}$, and the result remains the same. Then, we consider

$$F(z) = f(z)e^{2\pi i M z},$$

and note that F is bounded by 1 on the positive real and positive imaginary axes. Since we also have $|F(z)| \leq Ce^{c|z|}$ in the quadrant, we conclude by the Phragmén-Lindelöf theorem that $|F(z)| \leq 1$ for all z in Q, which implies $|f(z)| \leq e^{2\pi My}$. A similar argument for the other quadrants concludes Step 3 as well as the proof of the Paley-Wiener theorem.

We conclude with another version of the idea behind the Paley-Wiener theorem, this time characterizing the functions whose Fourier transform vanishes for all negative ξ.

Theorem 3.5 *Suppose f and $\hat{f}$ have moderate decrease. Then $\hat{f}(\xi) = 0$ for all $\xi < 0$ if and only if f can be extended to a continuous and bounded function in the closed upper half-plane $\{z = x + iy : y \geq 0\}$ with f holomorphic in the interior.*

Proof. First assume $\hat{f}(\xi) = 0$ for $\xi < 0$. By the Fourier inversion formula

$$f(x) = \int_0^\infty \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi\,,$$

and we can extend f when $z = x + iy$ with $y \geq 0$ by

$$f(z) = \int_0^\infty \hat{f}(\xi) e^{2\pi i z\xi}\, d\xi\,.$$

Notice that the integral converges and that

$$|f(z)| \leq A \int_0^\infty \frac{d\xi}{1 + \xi^2} < \infty\,,$$

which proves the boundedness of f. The uniform convergence of

$$f_n(z) = \int_0^n \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi$$

to $f(z)$ in the closed half-plane establishes the continuity of f there, and its holomorphicity in the interior.

For the converse, we argue in the spirit of the proof of Theorem 3.3. For ϵ and δ positive, we set

$$f_{\epsilon,\delta}(z) = \frac{f(z + i\delta)}{(1 - i\epsilon z)^2}.$$

Then $f_{\epsilon,\delta}$ is holomorphic in a region containing the closed upper half-plane. One also shows as before, using Cauchy's theorem, that $\widehat{f_{\epsilon,\delta}}(\xi) = 0$

for all $\xi < 0$. Then, passing to the limit successively, one has $\widehat{f_{\epsilon,0}}(\xi) = 0$ for $\xi < 0$, and finally $\hat{f}(\xi) = \widehat{f_{0,0}}(\xi) = 0$ for all $\xi < 0$.

Remark. The reader should note a certain analogy between the above theorem and Theorem 7.1 in Chapter 3. Here we are dealing with a function holomorphic in the upper half-plane, and there with a function holomorphic in a disc. In the present case the Fourier transform vanishes when $\xi < 0$, and in the earlier case, the Fourier coefficients vanish when $n < 0$.

4 Exercises

1. Suppose f is continuous and of moderate decrease, and $\hat{f}(\xi) = 0$ for all $\xi \in \mathbb{R}$. Show that $f = 0$ by completing the following outline:

(a) For each fixed real number t consider the two functions

$$A(z) = \int_{-\infty}^{t} f(x)e^{-2\pi iz(x-t)}\,dx \quad \text{and} \quad B(z) = -\int_{t}^{\infty} f(x)e^{-2\pi iz(x-t)}\,dx.$$

Show that $A(\xi) = B(\xi)$ for all $\xi \in \mathbb{R}$.

(b) Prove that the function F equal to A in the closed upper half-plane, and B in the lower half-plane, is entire and bounded, thus constant. In fact, show that $F = 0$.

(c) Deduce that

$$\int_{-\infty}^{t} f(x)\,dx = 0,$$

for all t, and conclude that $f = 0$.

2. If $f \in \mathfrak{F}_a$ with $a > 0$, then for any positive integer n one has $f^{(n)} \in \mathfrak{F}_b$ whenever $0 \leq b < a$.

[Hint: Modify the solution to Exercise 8 in Chapter 2.]

3. Show, by contour integration, that if $a > 0$ and $\xi \in \mathbb{R}$ then

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{a}{a^2 + x^2} e^{-2\pi ix\xi}\,dx = e^{-2\pi a|\xi|},$$

and check that

$$\int_{-\infty}^{\infty} e^{-2\pi a|\xi|}e^{2\pi i\xi x}\,d\xi = \frac{1}{\pi}\frac{a}{a^2 + x^2}.$$

4. Suppose Q is a polynomial of degree ≥ 2 with distinct roots, none lying on the real axis. Calculate

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x\xi}}{Q(x)}\, dx, \qquad \xi \in \mathbb{R}$$

in terms of the roots of Q. What happens when several roots coincide?

[Hint: Consider separately the cases $\xi < 0$, $\xi = 0$, and $\xi > 0$. Use residues.]

5. More generally, let $R(x) = P(x)/Q(x)$ be a rational function with (degree Q) $\geq$ (degree$P$)+2 and $Q(x) \neq 0$ on the real axis.

(a) Prove that if $\alpha_1, \ldots, \alpha_k$ are the roots of R in the upper half-plane, then there exists polynomials $P_j(\xi)$ of degree less than the multiplicity of α_j so that

$$\int_{-\infty}^{\infty} R(x)e^{-2\pi i x\xi}\, dx = \sum_{j=1}^{k} P_j(\xi)e^{-2\pi i \alpha_j \xi}, \qquad \text{when } \xi < 0.$$

(b) In particular, if $Q(z)$ has no zeros in the upper half-plane, then $\int_{-\infty}^{\infty} R(x)e^{-2\pi i x\xi}\, dx = 0$ for $\xi < 0$.

(c) Show that similar results hold in the case $\xi > 0$.

(d) Show that

$$\int_{-\infty}^{\infty} R(x)e^{-2\pi i x\xi}\, dx = O(e^{-a|\xi|}), \qquad \xi \in \mathbb{R}$$

as $|\xi| \to \infty$ for some $a > 0$. Determine the best possible a's in terms of the roots of R.

[Hint: For part (a), use residues. The powers of ξ appear when one differentiates the function $f(z) = R(z)e^{-2\pi i z\xi}$ (as in the formula of Theorem 1.4 in the previous chapter). For part (c) argue in the lower half-plane.]

6. Prove that

$$\frac{1}{\pi}\sum_{n=-\infty}^{\infty} \frac{a}{a^2 + n^2} = \sum_{n=-\infty}^{\infty} e^{-2\pi a|n|}$$

whenever $a > 0$. Hence show that the sum equals $\coth \pi a$.

7. The Poisson summation formula applied to specific examples often provides interesting identities.

(a) Let τ be fixed with $\mathrm{Im}(\tau) > 0$. Apply the Poisson summation formula to

$$f(z) = (\tau + z)^{-k},$$

where k is an integer ≥ 2, to obtain

$$\sum_{n=-\infty}^{\infty} \frac{1}{(\tau+n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} m^{k-1} e^{2\pi i m \tau}.$$

(b) Set $k = 2$ in the above formula to show that if $\operatorname{Im}(\tau) > 0$, then

$$\sum_{n=-\infty}^{\infty} \frac{1}{(\tau+n)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}.$$

(c) Can one conclude that the above formula holds true whenever τ is any complex number that is not an integer?

[Hint: For (a), use residues to prove that $\hat{f}(\xi) = 0$, if $\xi < 0$, and

$$\hat{f}(\xi) = \frac{(-2\pi i)^k}{(k-1)!} \xi^{k-1} e^{2\pi i \xi \tau}, \qquad \text{when } \xi > 0.]$$

8. Suppose $\hat{f}$ has compact support contained in $[-M, M]$ and let $f(z) = \sum_{n=0}^{\infty} a_n z^n$. Show that

$$a_n = \frac{(2\pi i)^n}{n!} \int_{-M}^{M} \hat{f}(\xi) \xi^n \, d\xi,$$

and as a result

$$\limsup_{n\to\infty} (n!|a_n|)^{1/n} \leq 2\pi M.$$

In the converse direction, let f be *any* power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ with $\limsup_{n\to\infty}(n!|a_n|)^{1/n} \leq 2\pi M$. Then, f is holomorphic in the complex plane, and for every $\epsilon > 0$ there exists $A_\epsilon > 0$ such that

$$|f(z)| \leq A_\epsilon e^{2\pi(M+\epsilon)|z|}.$$

9. Here are further results similar to the Phragmén-Lindelöf theorem.

(a) Let F be a holomorphic function in the right half-plane that extends continuously to the boundary, that is, the imaginary axis. Suppose that $|F(iy)| \leq 1$ for all $y \in \mathbb{R}$, and

$$|F(z)| \leq Ce^{c|z|^\gamma}$$

for some $c, C > 0$ and $\gamma < 1$. Prove that $|F(z)| \leq 1$ for all z in the right half-plane.

(b) More generally, let S be a sector whose vertex is the origin, and forming an angle of π/β. Let F be a holomorphic function in S that is continuous on the closure of S, so that $|F(z)| \leq 1$ on the boundary of S and

$$|F(z)| \leq Ce^{c|z|^\alpha} \text{ for all } z \in S$$

for some $c, C > 0$ and $0 < \alpha < \beta$. Prove that $|F(z)| \leq 1$ for all $z \in S$.

10. This exercise generalizes some of the properties of $e^{-\pi x^2}$ related to the fact that it is its own Fourier transform.

Suppose $f(z)$ is an entire function that satisfies

$$|f(x+iy)| \leq ce^{-ax^2+by^2}$$

for some $a, b, c > 0$. Let

$$\hat{f}(\zeta) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i x\zeta}\, dx.$$

Then, $\hat{f}$ is an entire function of ζ that satisfies

$$|\hat{f}(\xi+i\eta)| \leq c' e^{-a'\xi^2+b'\eta^2}$$

for some $a', b', c' > 0$.

[Hint: To prove $\hat{f}(\xi) = O(e^{-a'\xi^2})$, assume $\xi > 0$ and change the contour of integration to $x - iy$ for some $y > 0$ fixed, and $-\infty < x < \infty$. Then

$$\hat{f}(\xi) = O(e^{-2\pi y\xi}e^{by^2}).$$

Finally, choose $y = d\xi$ where d is a small constant.]

11. One can give a neater formulation of the result in Exercise 10 by proving the following fact.

Suppose $f(z)$ is an entire function of strict order 2, that is,

$$f(z) = O(e^{c_1|z|^2})$$

for some $c_1 > 0$. Suppose also that for x real,

$$f(x) = O(e^{-c_2|x|^2})$$

for some $c_2 > 0$. Then

$$|f(x+iy)| = O(e^{-ax^2+by^2})$$

for some $a, b > 0$. The converse is obviously true.

12. The principle that a function and its Fourier transform cannot both be too small at infinity is illustrated by the following theorem of Hardy.

If f is a function on $\mathbb{R}$ that satisfies

$$f(x) = O(e^{-\pi x^2}) \qquad \text{and} \qquad \hat{f}(\xi) = O(e^{-\pi \xi^2}),$$

then f is a constant multiple of $e^{-\pi x^2}$. As a result, if $f(x) = O(e^{-\pi A x^2})$, and $\hat{f}(\xi) = O(e^{-\pi B \xi^2})$, with $AB > 1$ and $A, B > 0$, then f is identically zero.

(a) If f is even, show that $\hat{f}$ extends to an even entire function. Moreover, if $g(z) = \hat{f}(z^{1/2})$, then g satisfies

$$|g(x)| \leq ce^{-\pi x} \qquad \text{and} \qquad |g(z)| \leq ce^{\pi R \sin^2(\theta/2)} \leq ce^{\pi |z|}$$

when $x \in \mathbb{R}$ and $z = Re^{i\theta}$ with $R \geq 0$ and $\theta \in \mathbb{R}$.

(b) Apply the Phragmén-Lindelöf principle to the function

$$F(z) = g(z)e^{\gamma z} \qquad \text{where } \gamma = i\pi \frac{e^{-i\pi/(2\beta)}}{\sin \pi/(2\beta)}$$

and the sector $0 \leq \theta \leq \pi/\beta < \pi$, and let $\beta \to \pi$ to deduce that $e^{\pi z} g(z)$ is bounded in the closed upper half-plane. The same result holds in the lower half-plane, so by Liouville's theorem $e^{\pi z} g(z)$ is constant, as desired.

(c) If f is odd, then $\hat{f}(0) = 0$, and apply the above argument to $\hat{f}(z)/z$ to deduce that $f = \hat{f} = 0$. Finally, write an arbitrary f as an appropriate sum of an even function and an odd function.

5 Problems

1. Suppose $\hat{f}(\xi) = O(e^{-a|\xi|^p})$ as $|\xi| \to \infty$, for some $p > 1$. Then f is holomorphic for all z and satisfies the growth condition

$$|f(z)| \leq Ae^{a|z|^q}$$

where $1/p + 1/q = 1$.

Note that on the one hand, when $p \to \infty$ then $q \to 1$, and this limiting case can be interpreted as part of Theorem 3.3. On the other hand, when $p \to 1$ then $q \to \infty$, and this limiting case in a sense brings us back to Theorem 2.1.

[Hint: To prove the result, use the inequality $-\xi^p + \xi u \leq u^q$, which is valid when ξ and u are non-negative. To establish this inequality, examine separately the cases $\xi^p \geq \xi u$ and $\xi^p < \xi u$; note also that the functions $\xi = u^{q-1}$ and $u = \xi^{p-1}$ are inverses of each other because $(p-1)(q-1) = 1$.]